



NSW Education Standards Authority

2018 HIGHER SCHOOL CERTIFICATE EXAMINATION

Physics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Draw diagrams using pencil
- Calculators approved by NESA may be used
- A data sheet, formulae sheets and Periodic Table are provided at the back of this paper

Total marks: 100

Section I – 75 marks (pages 2–32)

This section has two parts, Part A and Part B

Part A – 20 marks

- Attempt Questions 1–20
- Allow about 35 minutes for this part

Part B – 55 marks

- Attempt Questions 21–30
- Allow about 1 hour and 40 minutes for this part

Section II – 25 marks (pages 33–43)

- Attempt ONE question from Questions 31–35
- Allow about 45 minutes for this section

Section I
75 marks

Part A – 20 marks

Attempt Questions 1–20

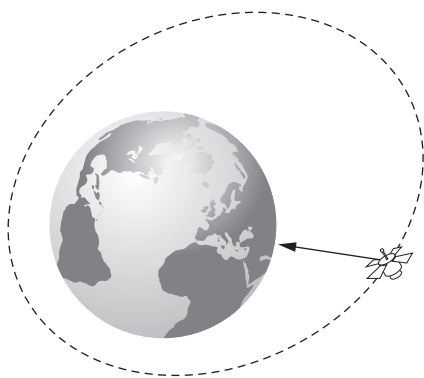
Allow about 35 minutes for this part

Use the multiple-choice answer sheet for Questions 1–20.

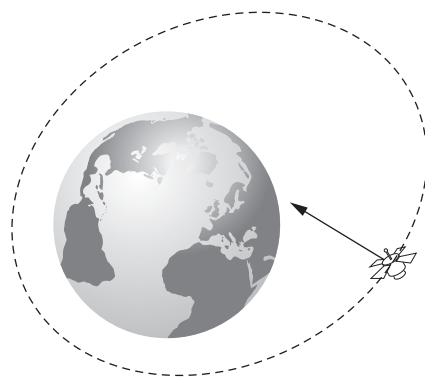
1 A satellite orbits Earth as shown.

Which diagram correctly shows the direction of the satellite's acceleration?

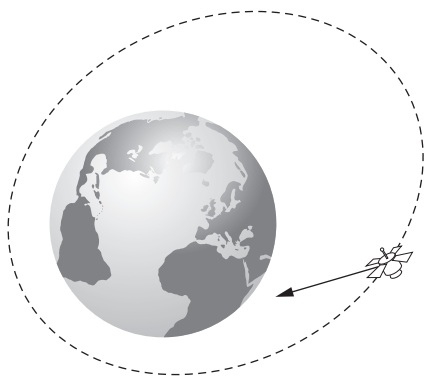
A.



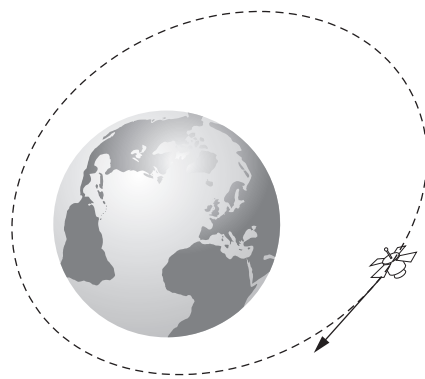
B.



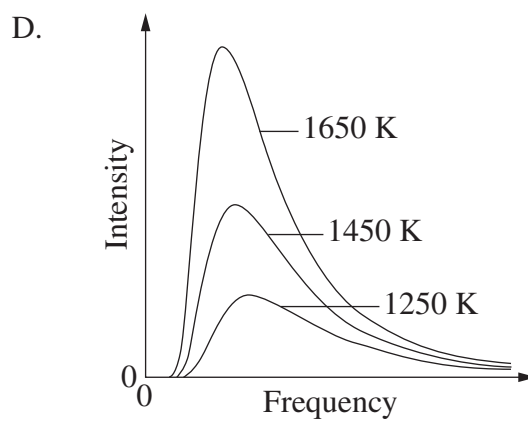
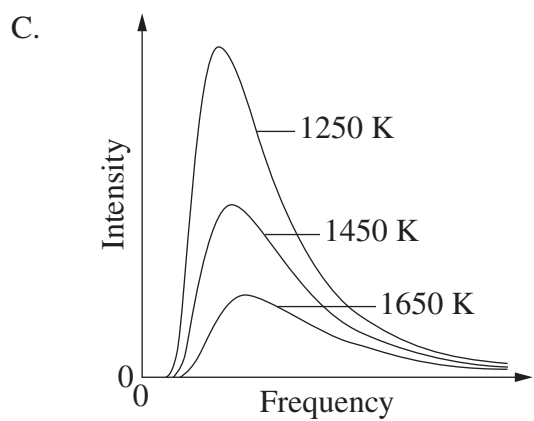
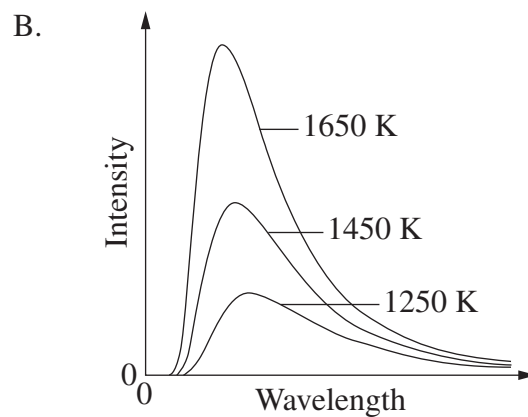
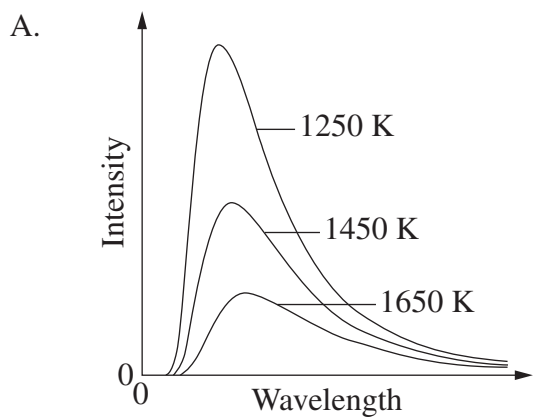
C.



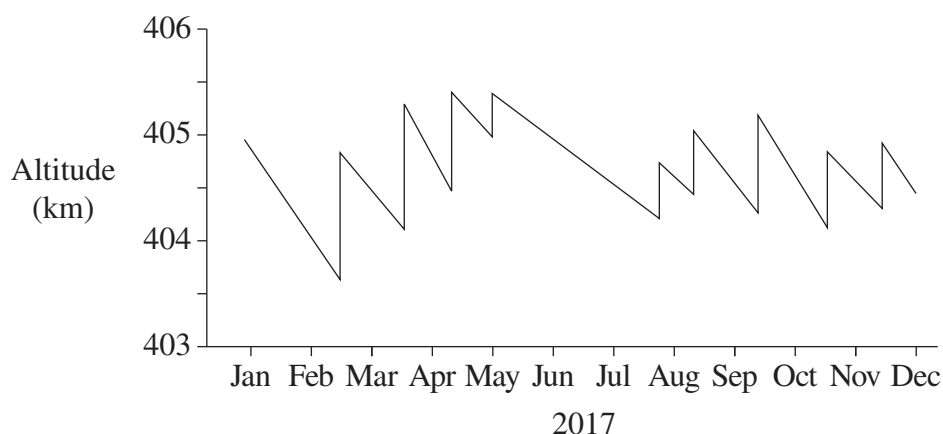
D.



- 2 Which graph is consistent with predictions resulting from Planck's hypothesis regarding radiation from hot objects?



- 3 The graph shows the altitude of the International Space Station (ISS) during 2017.



The altitude can only be boosted by supply craft visiting the ISS.

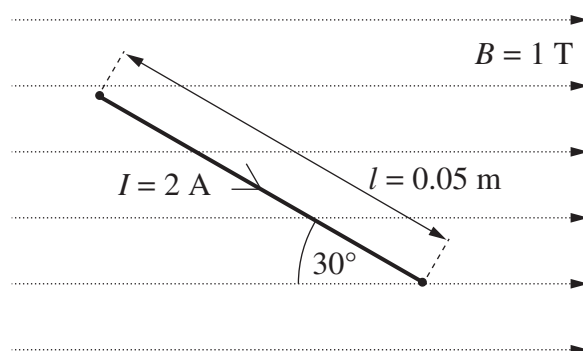
Why does the altitude decrease in the times between height boosts?

- A. Momentum of the ISS is being transferred to air molecules.
 - B. The moon's gravity changes the net force on the ISS as it orbits Earth.
 - C. The decrease in altitude makes it possible for a supply craft to reach the ISS.
 - D. The total mass of the ISS changes with the deliveries from each supply craft.
- 4 A motor, battery and ammeter are connected in series. When the motor is turning at full speed, the ammeter has a reading of 0.1 A. While the motor is spinning, a person holds the shaft of the motor to stop it.

Which row of the table correctly identifies the change in the ammeter reading and an explanation for the change?

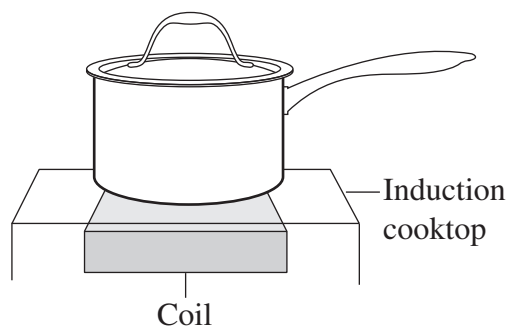
	<i>Reading on ammeter</i>	<i>Explanation</i>
A.	Decreases	Decrease in back emf
B.	Increases	Increase in back emf
C.	Decreases	Increase in back emf
D.	Increases	Decrease in back emf

- 5 The diagram shows a current-carrying conductor in a magnetic field.



What is the magnitude of the force on the conductor?

- A. 0 N
 - B. 0.05 N
 - C. 0.09 N
 - D. 0.10 N
- 6 The diagram shows a saucepan of water on an induction cooktop.



Which row of the table correctly identifies a property of the material used to make the saucepan and the frequency of the changing magnetic field produced by the coil?

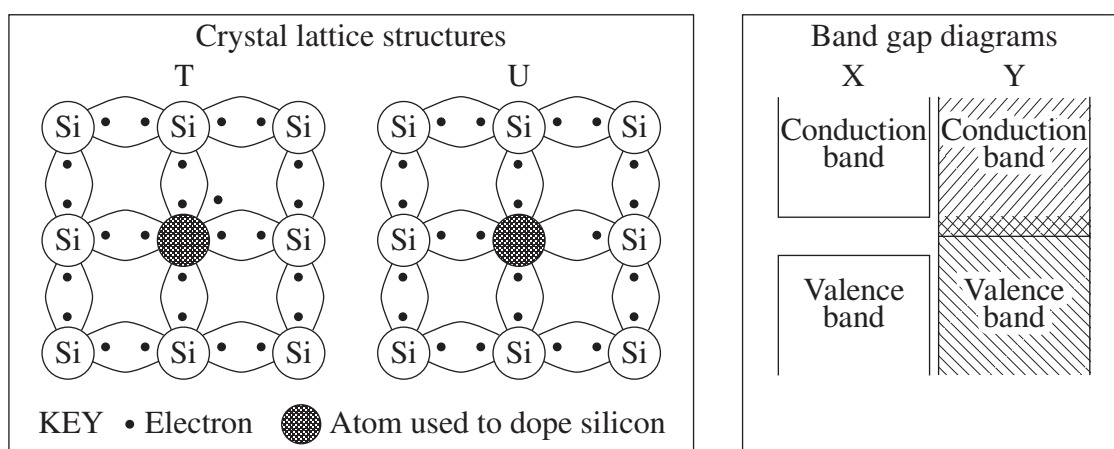
	<i>Property of saucepan</i>	<i>Frequency</i>
A.	Insulator	High (50 kHz)
B.	Conductor	High (50 kHz)
C.	Insulator	Low (50 Hz)
D.	Conductor	Low (50 Hz)

- 7 A planet X has twice the mass and twice the radius of Earth.

What is the magnitude of the gravitational acceleration close to the surface of planet X ?

- A. $\frac{1}{2}g$
 B. $1g$
 C. $2g$
 D. $4g$

- 8 Crystal lattice structures for two types of doped semiconductors and band gap diagrams for two types of materials are shown.



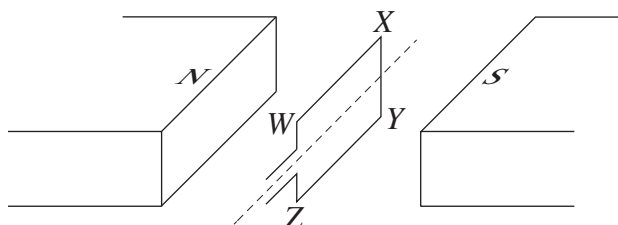
Which row of the table correctly matches the type of doped semiconductor with its crystal lattice structure and band gap diagram?

	<i>Type of semiconductor</i>	<i>Crystal lattice structure</i>	<i>Band gap diagram</i>
A.	<i>n</i> -type	U	X
B.	<i>n</i> -type	T	Y
C.	<i>p</i> -type	T	Y
D.	<i>p</i> -type	U	X

- 9 A hypothetical journey to a distant star might be accomplished in an astronaut's life span by travelling at relativistic speeds.

What is the key concept which underpins such a hypothetical journey?

- A. From the frame of reference of the spacecraft making the journey, the distance to the star is less.
 - B. Clocks on the spacecraft run slower and hence the rate at which fuel is used for the journey is decreased.
 - C. Relativistic effects on the spacecraft reduce its mass, making it possible to accelerate to the speeds needed for such a journey.
 - D. The relativistic increase in the mass of the fuel on board makes it possible to complete a longer journey with less fuel.
- 10 The diagram shows some parts of a simple DC motor.



Which row of the table correctly describes the direction of force acting on side WX and the direction of torque this produces on the coil?

	<i>Direction of force acting on WX</i>	<i>Direction of torque produced on the coil by the force acting on WX</i>
A.	Remains constant	Remains constant
B.	Remains constant	Reverses every 180°
C.	Reverses every 180°	Remains constant
D.	Reverses every 180°	Reverses every 180°

- 11 During the launch of a space vehicle from Earth, an astronaut feels an increased downward g force.

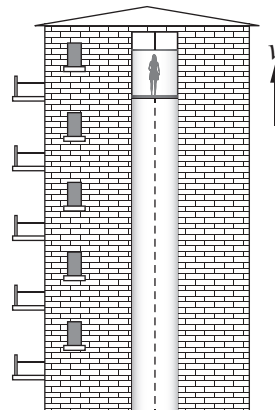
In which of the following situations would a person also feel an increased downward g force?

A.



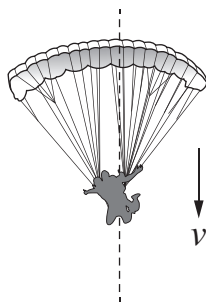
Roller-coaster speeding up

B.



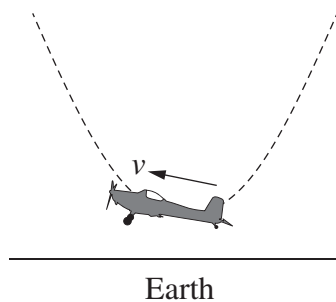
Lift slowing down

C.



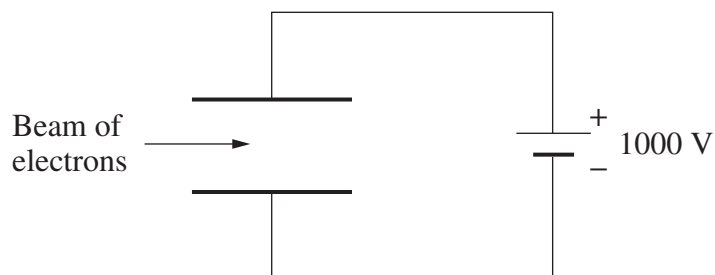
Parachute falling at a constant velocity

D.



Plane pulling out of a dive

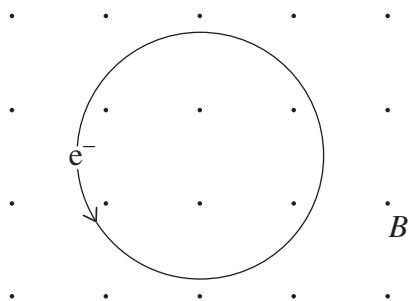
- 12 The diagram shows electrons travelling in a vacuum at $2 \times 10^6 \text{ m s}^{-1}$ between two charged metal plates $1 \times 10^{-3} \text{ m}$ apart.



A magnetic field is to be applied to make the electrons continue to travel in a straight line.

What is the magnitude and direction of the magnetic field that is to be applied?

- A. $5 \times 10^{-1} \text{ T}$ into the page
 - B. $5 \times 10^{-1} \text{ T}$ out of the page
 - C. $1 \times 10^6 \text{ T}$ into the page
 - D. $1 \times 10^6 \text{ T}$ out of the page
- 13 An electron moves in a circular path with radius r in a magnetic field as shown.



If the speed of the electron is increased, which row of the table correctly shows the effects of this change?

	<i>Force on electron</i>	<i>Radius of path</i>
A.	Increases	Decreases
B.	Increases	Increases
C.	Decreases	Decreases
D.	Decreases	Increases

- 14 A pendulum can be used to determine the acceleration due to gravity using the relationship

$$T = 2\pi\sqrt{\frac{l}{g}}$$

where T is the period and l is the length of the pendulum.

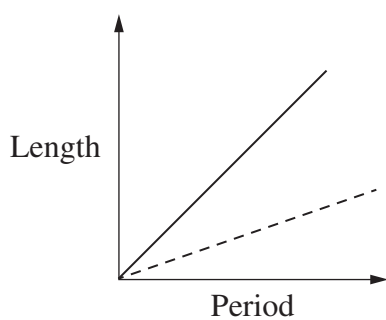
The acceleration due to gravity on the surface of Mars is less than that on Earth.

Which graph relates the variables for the pendulum correctly for both planets?

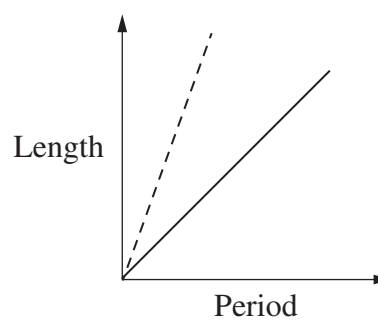
KEY

— Earth data
- - - Mars data

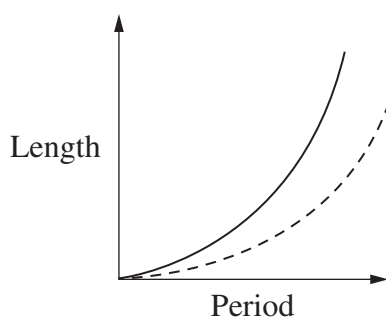
A.



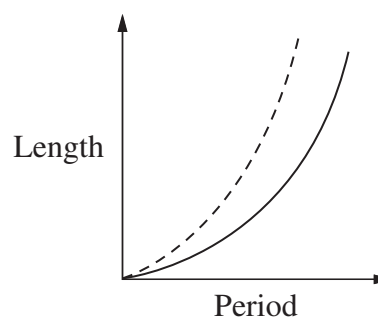
B.



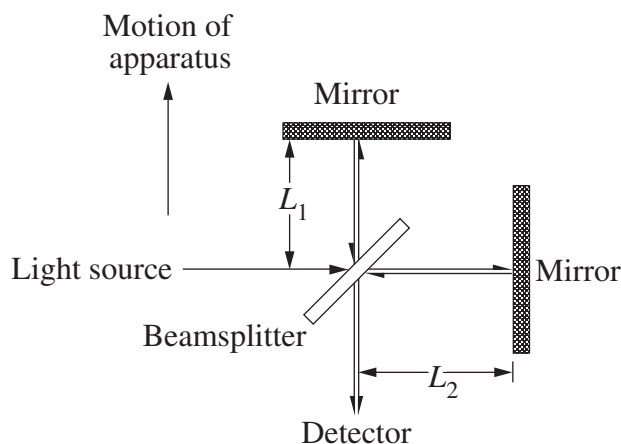
C.



D.



- 15 The diagram shows a simplified model of the Michelson–Morley experiment. It can be assumed that distances L_1 and L_2 are equal without affecting the outcome.

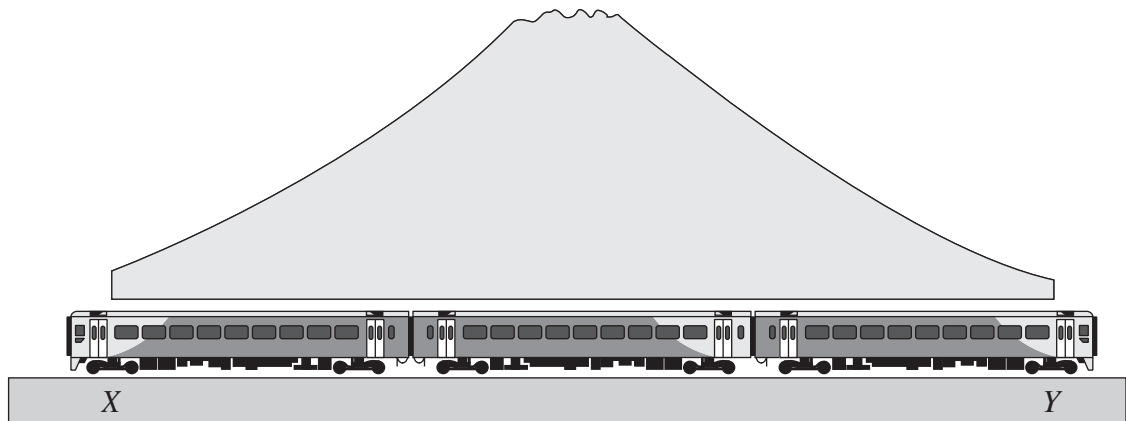


The times taken for the light to travel in both directions along the lengths L_1 and L_2 are t_1 and t_2 respectively.

What was Michelson attempting to demonstrate in this experiment?

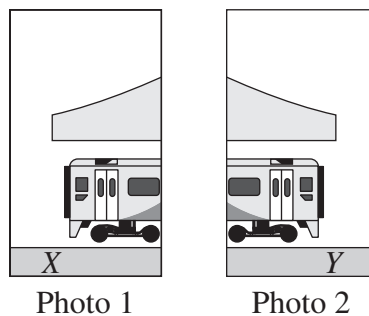
- A. The relativistic contraction of L_1 would cause t_1 to be less than t_2 .
- B. The ether is carried through space with Earth, which would cause t_1 to be equal to t_2 .
- C. The times t_1 and t_2 would be the same because the velocity of the apparatus was much less than the speed of light.
- D. The motion of the apparatus resulting from Earth's orbital motion around the sun would cause t_1 to be greater than t_2 .

- 16 When a train is at rest in a tunnel, the train is slightly longer than the tunnel.



In a thought experiment, the train is travelling from left to right fast enough relative to the tunnel that its length contracts and it fits inside the tunnel.

An observer on the ground sets up two cameras, at X and Y, to take photos at exactly the same time. The photos show that both ends of the train are inside the tunnel.

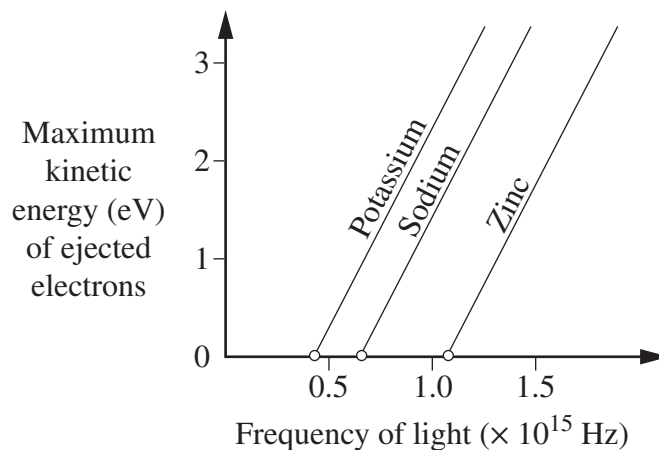


A passenger travelling on the train at its centre can see both ends of the tunnel and is later shown the photos.

From the point of view of the passenger, what is observed and what can be deduced about the photos?

- A. The tunnel's length contracts so the train does not fit, and photo 2 is taken before photo 1.
- B. The tunnel's length contracts so the train does not fit, and photos 1 and 2 are taken at the same time.
- C. The tunnel appears to expand due to length contraction of the train, allowing it to fit in the tunnel, and photo 1 is taken before photo 2.
- D. The tunnel appears to expand due to length contraction of the train, allowing it to fit in the tunnel, and photos 1 and 2 are taken at the same time.

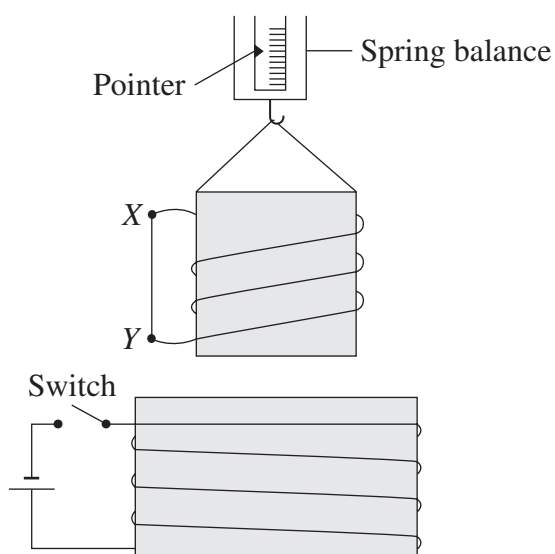
- 17 The graph shows the maximum kinetic energy of electrons ejected from different metals as a function of the frequency of the incident light.



What can be deduced from this graph?

- A. The maximum kinetic energy of ejected electrons is proportional to the number of photons incident on the metal surface.
- B. More photons are required to cause an electron to be ejected from zinc than from potassium.
- C. Any photon that can eject an electron from the surface of zinc must also be able to cause an electron to be ejected from potassium.
- D. For any given frequency that causes electrons to be ejected from all three metals, the number of electrons ejected is always greatest for potassium.

- 18 An experiment is set up as shown.

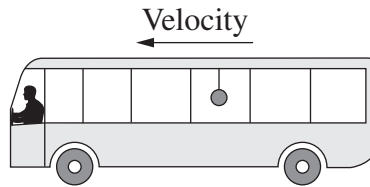


When the switch is closed, the reading on the spring balance changes immediately, then returns to the initial reading.

Which row of the table correctly shows the direction of the current through the straight conductor XY and the direction in which the pointer on the spring balance initially moves?

	<i>Direction of current through the straight conductor</i>	<i>Direction in which the pointer initially moves</i>
A.	From X to Y	Down
B.	From X to Y	Up
C.	From Y to X	Down
D.	From Y to X	Up

- 19 A mass was hanging from the roof of a bus that was travelling forward on a horizontal road at a constant velocity.



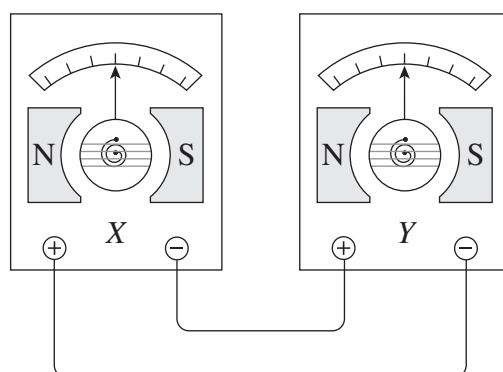
The string holding the mass was cut. At the same instant, the bus driver applied the brakes, causing the bus to slow down at a rate of 3 m s^{-2} .

To an observer outside the bus, the mass follows a parabolic trajectory.

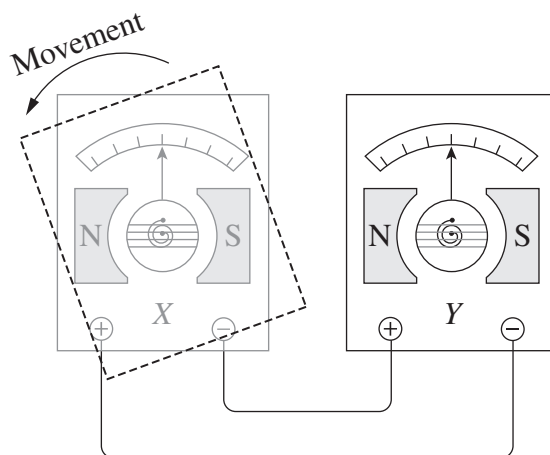
Which statement correctly describes the resulting motion of the mass observed from within the frame of reference of the moving bus?

- A. The mass travelled in a straight line vertically downwards.
- B. The mass travelled in a straight line downwards and towards the front of the bus.
- C. The mass travelled in a parabolic path downwards and towards the back of the bus.
- D. The mass travelled in a parabolic path downwards and towards the front of the bus.

- 20 Two identical galvanometers X and Y were connected as shown and placed on a desk.



The meter X was then picked up and rotated suddenly anticlockwise to the position shown by the dotted outline, while the meter Y remained stationary on the desk.



When this action is carried out, the pointers on both meters move relative to the scales on the meters.

Which row of the table correctly identifies the movement of the pointer on Y and the reason for its behaviour?

	<i>Movement of pointer</i>	<i>Reason</i>
A.	Towards right	The movement of X produces a current that flows into the opposite terminal of Y .
B.	Towards right	According to Lenz's law the effect produced must oppose the cause of the flux.
C.	Towards left	Y behaves as a motor with the current flowing through it from right to left.
D.	Towards left	For every force, there is an equal and opposite force.

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Centre Number

Physics

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Student Number

Section I Part B Answer Booklet

55 marks

Attempt Questions 21–30

Allow about 1 hour and 40 minutes for this part

Instructions

- Write your Centre Number and Student Number at the top of this page.
- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Show all relevant working in questions involving calculations.
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.

Please turn over

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Question 21 (4 marks)

- (a) Compare the force of gravity exerted on the moon by Earth with the force of gravity exerted on Earth by the moon. **2**

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- (b) The acceleration due to gravity on the moon is 1.6 m s^{-2} and on Earth it is 9.8 m s^{-2} . Quantitatively compare the mass and weight of a 70 kg person on the moon and on Earth. **2**

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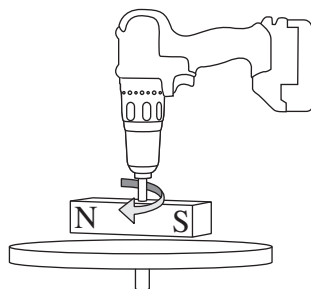
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Please turn over

Question 22 (6 marks)

- (a) A drill spins a magnet above a non-magnetic metal disc which is free to rotate.

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Explain the effect of the rotating magnet on the disc.

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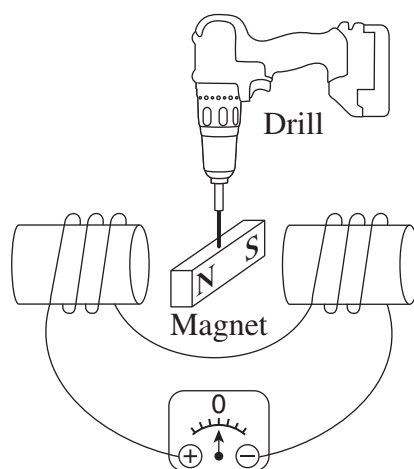
Question 22 continues on page 21

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Question 22 (continued)

- (b) The diagram shows a magnet attached to an electric drill so that it can be rotated between two coils connected to a voltmeter.

3



The drill starts from rest and gradually speeds up, reaching its full speed after three revolutions.

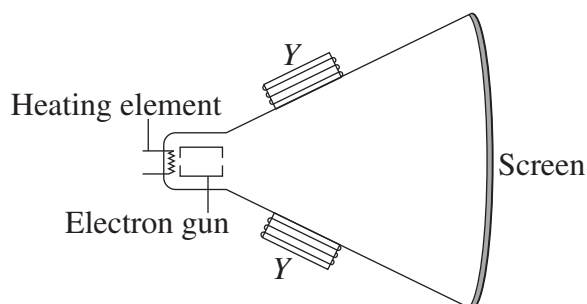
Sketch a graph showing the induced emf across the coils during the time that it takes the magnet to reach its full speed.



End of Question 22

Question 23 (5 marks)

The diagram shows a cathode ray tube in a television.



- (a) Outline energy changes associated with the electrons passing through the gun, and when they strike the screen. 2

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- (b) Explain the role of the components labelled *Y* in forming an image on the screen. 3

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Question 24 (5 marks)

- (a) Outline TWO features of high-voltage transmission lines that contribute to the safe transmission of electricity. 2

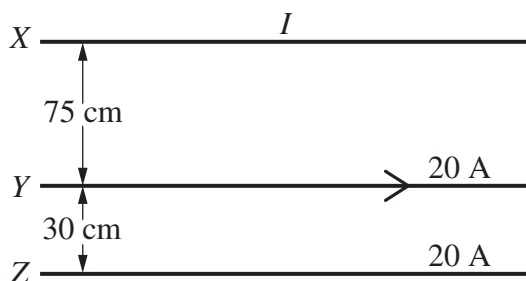
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- (b) Three parallel wires X, Y and Z all carry electric currents. A force of attraction is produced between Y and Z. There is zero net force on Y. 3



What is the magnitude and direction of the current in X?

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Question 25 (6 marks)

- (a) State ONE benefit and ONE limitation of the use of superconductors in the transmission of electricity through power grids. 2

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- (b) Contrast features of electrical conduction in a pure metal at room temperature with electrical conduction in a pure metallic superconductor below its critical temperature. 4

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Question 26 (4 marks)

Outline the similarities and differences between the effects of electric fields and gravitational fields on matter. In your answer, refer to the definitions of these fields.

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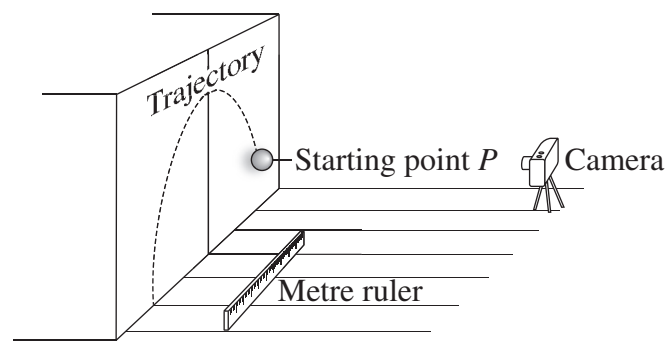
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Question 27 (6 marks)

- (a) The diagram shows a camera and a ruler set up to obtain data about a projectile's motion along the trajectory shown. The entire trajectory is visible through the camera.

3



Identify ONE of the errors in this set-up and describe the effect of this error on the results.

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Question 27 continues on page 27

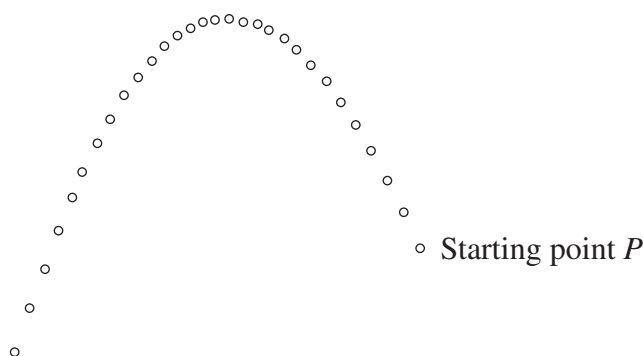
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Question 27 (continued)

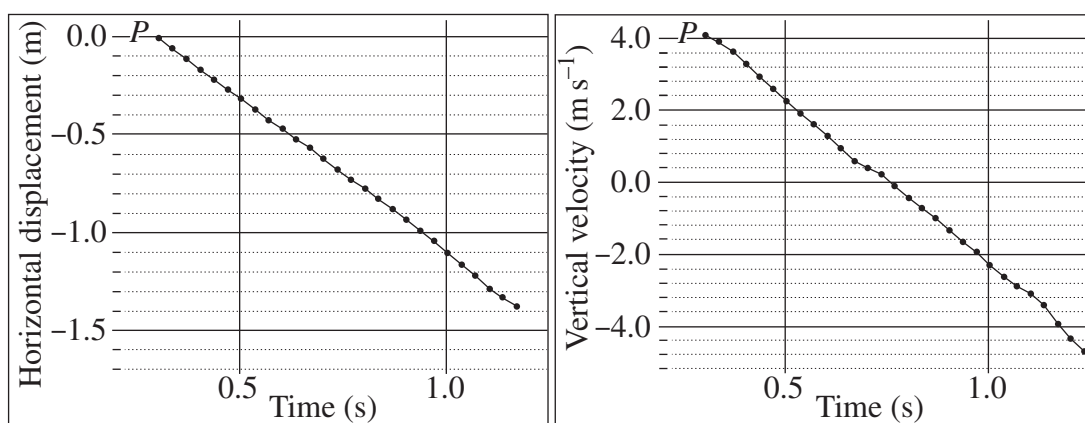
- (b) An experiment was set up based on the method described in part (a), but conducted so that the data obtained were valid.

3

The image shows the trajectory of the ball.



The graphs show data from this experiment.



Using the graphs, describe the velocity and acceleration of the ball quantitatively and qualitatively.

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End of Question 27

Question 28 (5 marks)

The radius of the moon is 1740 km. The moon's mass is 7.35×10^{22} kg. In this question, ignore the moon's rotational and orbital motion.

A 20 kg mass is launched vertically from the moon's surface at a velocity of 1200 m s^{-1} .

- (a) Show that the change in potential energy of the mass in moving from the surface to an altitude of 500 km is $1.26 \times 10^7 \text{ J}$. 2

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- (b) Calculate the velocity of the 20 kg mass at an altitude of 500 km. 3

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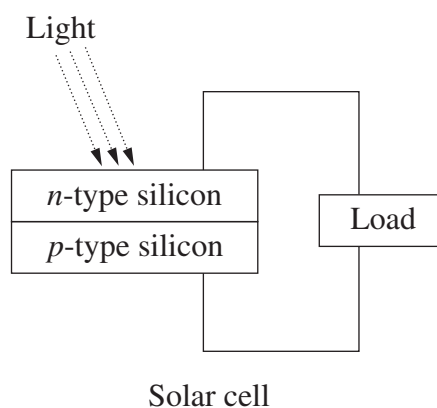
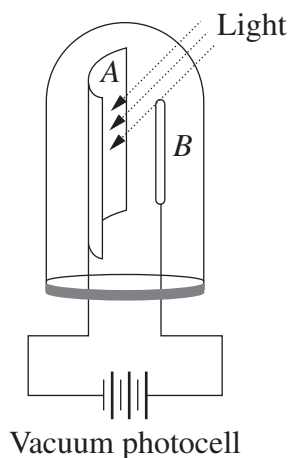
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Question 29 (8 marks)

The diagram shows two different cells.



- (a) How does light contribute to the flow of current in each of these two cells?

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- (b) Explain how ONE advance in scientific understanding and ONE advance in technology have led to the use of semiconductors.

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[illegible]

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Section I Part B extra writing space

If you use this space, clearly indicate which question you are answering.

Section I Part B extra writing space

If you use this space, clearly indicate which question you are answering.

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Physics

Section II

25 marks

Attempt ONE question from Questions 31–35

Allow about 45 minutes for this section

Answer parts (a)–(e) of one question in the Section II Writing Booklet. Extra writing booklets are available.

Show all relevant working in questions involving calculations.

	Pages
Question 31 Geophysics	34
Question 32 Medical Physics	35–36
Question 33 Astrophysics	37–38
Question 34 From Quanta to Quarks	39–40
Question 35 The Age of Silicon	41–43

Question 31 — Geophysics (25 marks)

Answer parts (a), (b) and (c) of the question on pages 2–4 of the Section II Writing Booklet. Start each part of the question on a new page.

- | | | | |
|-----|------|---|---|
| (a) | (i) | Seismic methods are used to investigate physical properties of Earth. Identify TWO other methods used by geophysicists. | 2 |
| | (ii) | How does a geophysical method, other than a seismic method, provide information about physical properties of Earth? | 3 |
| (b) | | In November 2017, a new polar-orbiting weather satellite NOAA–20 was placed in orbit. Its orbital period is 100 minutes and it orbits Earth at an altitude of 870 km. The average radius of Earth is 6370 km. | |
| | (i) | Explain ONE feature of NOAA–20’s orbit that will make it better for monitoring climate than a geostationary satellite. | 2 |
| | (ii) | Calculate the mass of Earth using the information provided about NOAA–20’s orbit. The answer is different to that shown on the data sheet provided. | 3 |
| (c) | | Explain how both Richer and Newton independently used data obtained from experiments using pendulums to determine the shape of Earth. | 4 |

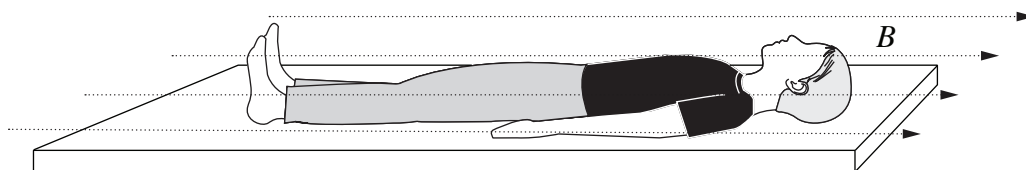
Answer parts (d) and (e) of the question on pages 6–8 of the Section II Writing Booklet. Start each part of the question on a new page.

- | | | | |
|-----|--|--|---|
| (d) | | How are seismic data used to provide a model of Earth’s core? Support your answer with a diagram. | 4 |
| (e) | | Explain the role of palaeomagnetic data in providing evidence that supports the theory of plate tectonics. | 7 |

Question 32 — Medical Physics (25 marks)

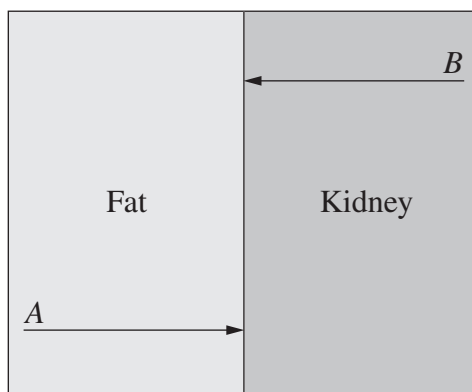
Answer parts (a), (b) and (c) of the question on pages 2–4 of the Section II Writing Booklet. Start each part of the question on a new page.

- (a) (i) Identify TWO specific uses of magnetic resonance imaging (MRI). 2
- (ii) The diagram shows a person in the magnetic field within an MRI machine. 3



Describe the effects of the application of the strong magnetic field, B , on hydrogen nuclei in the person's body.

- (b) The diagram represents fat and kidney tissues inside a human body. The acoustic impedances of these tissues are not the same.



- (i) Ultrasound waves A and B have the same frequency. 2

Compare the behaviour of ultrasound waves A and B at the fat–kidney boundary.

- (ii) The density of kidney tissue is 1050 kg m^{-3} and the velocity of ultrasound waves in kidney tissue is 1560 m s^{-1} . At the fat–kidney boundary, the intensity of the reflected ultrasound is 1.0 % of that of the incident ultrasound. 3

Calculate the acoustic impedance of fat tissue.

Question 32 continues on page 36

Question 32 (continued)

- (c) Iodine-124 is a radioactive element. The equation for its decay is shown. The symbol e^+ is sometimes shown as β^+ . 4



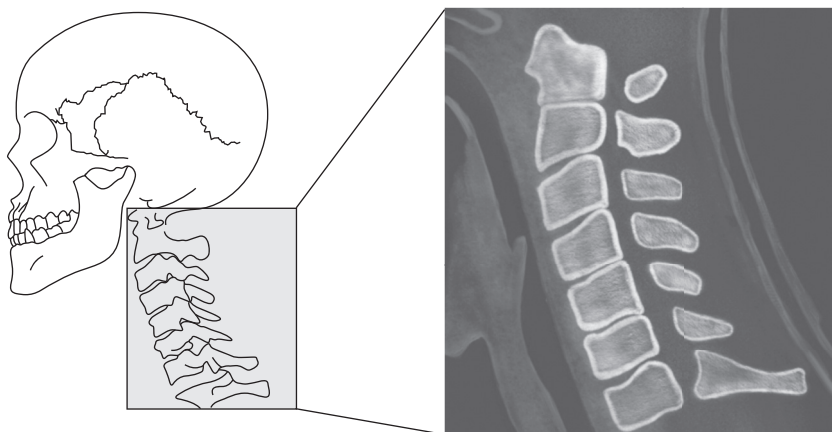
The following is a decay table for iodine-124.

<i>Time (h)</i>	0	20	40	60	80	100	120	140
<i>Proportion remaining</i>	1.000	0.871	0.758	0.661	0.575	0.501	0.436	0.355

Justify the use of iodine-124 as a medical diagnostic tool. Support your answer quantitatively.

Answer parts (d) and (e) of the question on pages 6–8 of the Section II Writing Booklet. Start each part of the question on a new page.

- (d) Explain how the Doppler effect is used for medical imaging. 4
- (e) The following image of the bones in a person's neck was produced using computed axial tomography (CAT). 7



Compare both the production and benefits of this CAT image with a conventional X-ray image of the same part of the person's body.

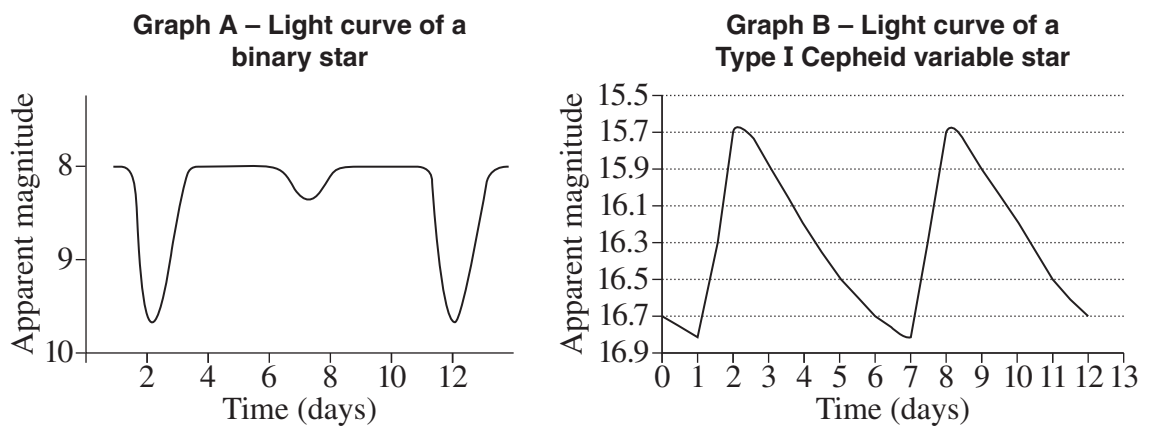
End of Question 32

Question 33 — Astrophysics (25 marks)

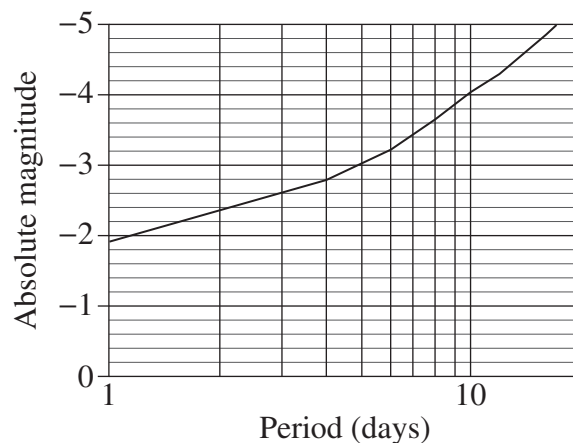
Answer parts (a), (b) and (c) of the question on pages 2–4 of the Section II Writing Booklet. Start each part of the question on a new page.

- (a) (i) Define the parsec with the aid of a diagram. 2
- (ii) Outline TWO limitations of using ground-based instruments to make trigonometric parallax measurements. 3

- (b) The diagrams show light curves of two variable stars.



- (i) Contrast the reasons for the variation in brightness of these two stars. 2
- (ii) A period–luminosity graph for Type I Cepheid variable stars is shown. 3



The average apparent magnitude of the Type I Cepheid variable star shown in Graph B is 16.3.

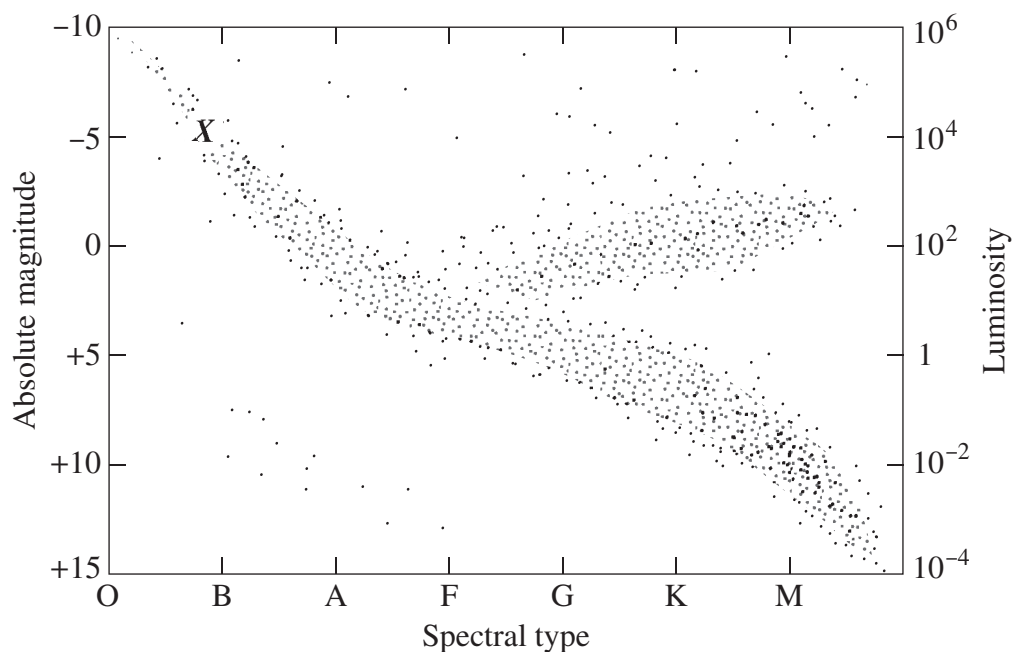
Calculate the distance of this Type I Cepheid variable star from Earth.

Question 33 continues on page 38

Question 33 (continued)

- (c) A H–R diagram is shown. Star X is a main sequence star of 12 solar masses.

4



Describe how star X will change physically and chemically as it continues to evolve.

Answer parts (d) and (e) of the question on pages 6–8 of the Section II Writing Booklet. Start each part of the question on a new page.

- (d) Explain how measurements made using photoelectric technologies have increased the understanding of celestial objects beyond that provided using photographic methods.
- (e) Explain how black body spectra, emission spectra and absorption spectra are used to analyse the surface temperature and chemical composition of stars. In your answer, refer to observations that can be made in a school laboratory.

4

7

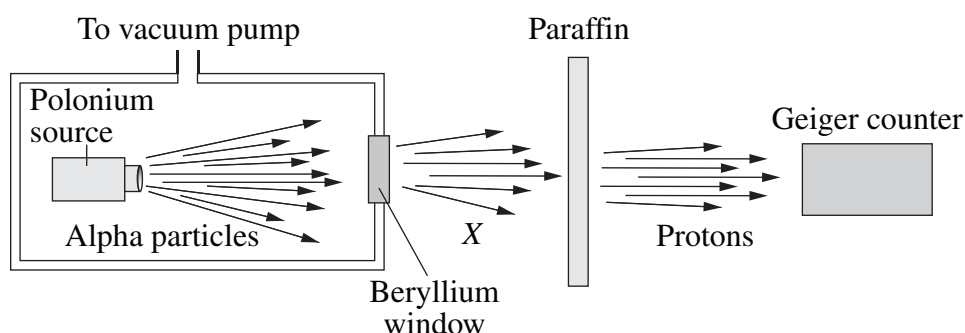
End of Question 33

Question 34 — From Quanta to Quarks (25 marks)

Answer parts (a), (b) and (c) of the question on pages 2–4 of the Section II Writing Booklet. Start each part of the question on a new page.

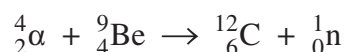
- (a) (i) Identify TWO limitations of the Bohr model of the atom. 2
- (ii) How does the Bohr model of the atom account for the emission spectrum of hydrogen? 3

- (b) (i) The diagram shows apparatus used to investigate subatomic particles. 2



How did Chadwick use a law of physics to identify a property of X?

- (ii) The following is a nuclear reaction that produces a neutron. 3



The table shows the masses of the particles in the reaction.

Particle	Mass (u)
${}^4_2\alpha$	4.0012
${}^9_4\text{Be}$	9.0122
${}^{12}_6\text{C}$	12.0000
${}^1_0\text{n}$	1.0087

Using the data from the table, calculate the energy released in this reaction. State your answer in joules.

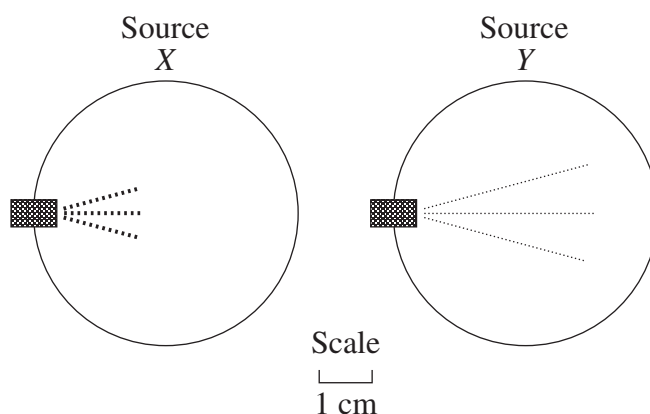
- (c) How did Pauli infer the existence and properties of a new particle from measurements of the energies of beta particles? Support your answer with a graph. 4

Question 34 continues on page 40

Question 34 (continued)

Answer parts (d) and (e) of the question on pages 6–8 of the Section II Writing Booklet. Start each part of the question on a new page.

- (d) Radioactive sources, *X* and *Y*, producing two different types of radiation were placed in identical Wilson cloud chambers. The diagram shows the trails produced by each. 4



Describe the changes in the nuclei of *X* and *Y* which have produced these trails.

- (e) Using the standard model, analyse the roles of both forces and particles in the current understanding of the atom. 7

End of Question 34

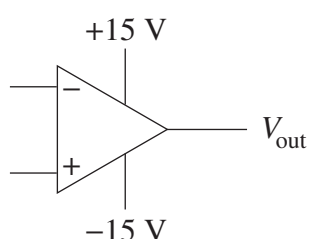
Question 35 — The Age of Silicon (25 marks)

Answer parts (a), (b) and (c) of the question on pages 2–4 of the Section II Writing Booklet. Start each part of the question on a new page.

- (a) (i) Outline TWO differences between an electric circuit and an electronic circuit. **2**

- (ii) How do the properties of silica make it an ideal material for use in electronic communication? **3**

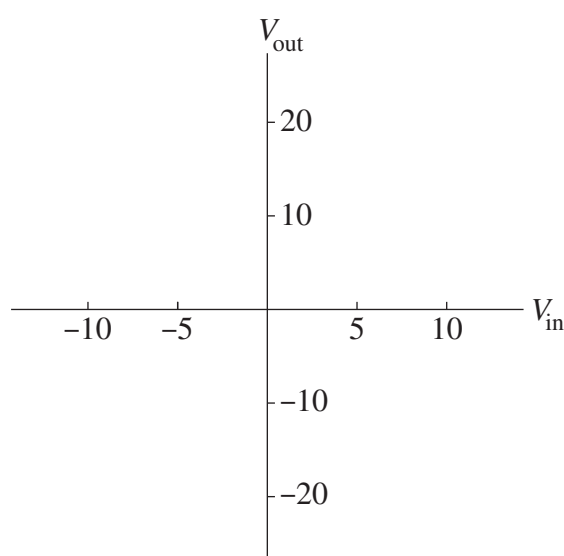
- (b) The diagram shows an operational amplifier connected to a power supply.



This operational amplifier is used to make an inverting amplifier by adding an input resistor of $27\text{ k}\Omega$ and a feedback resistor of $82\text{ k}\Omega$.

- (i) In your answer booklet, draw the inverting amplifier described. **2**

- (ii) Copy the axes shown into your answer booklet. Draw a graph showing the transfer characteristics of this inverting amplifier for the input range of -10 V to $+10\text{ V}$. **3**

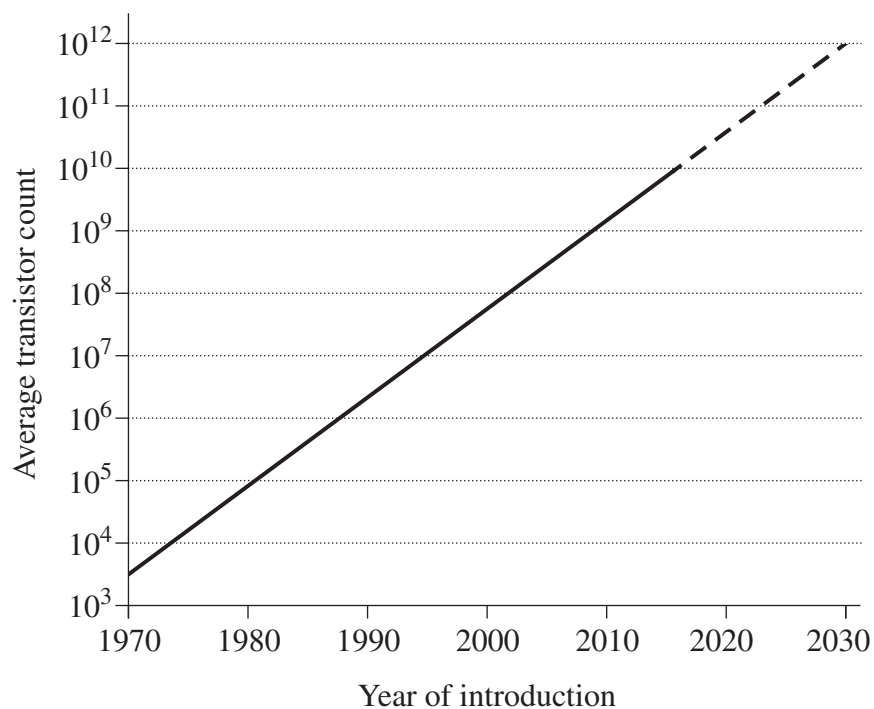


Question 35 continues on page 42

Question 35 (continued)

- (c) The graph shows the average number of transistors incorporated into integrated circuit chips from 1970 to 2015.

4



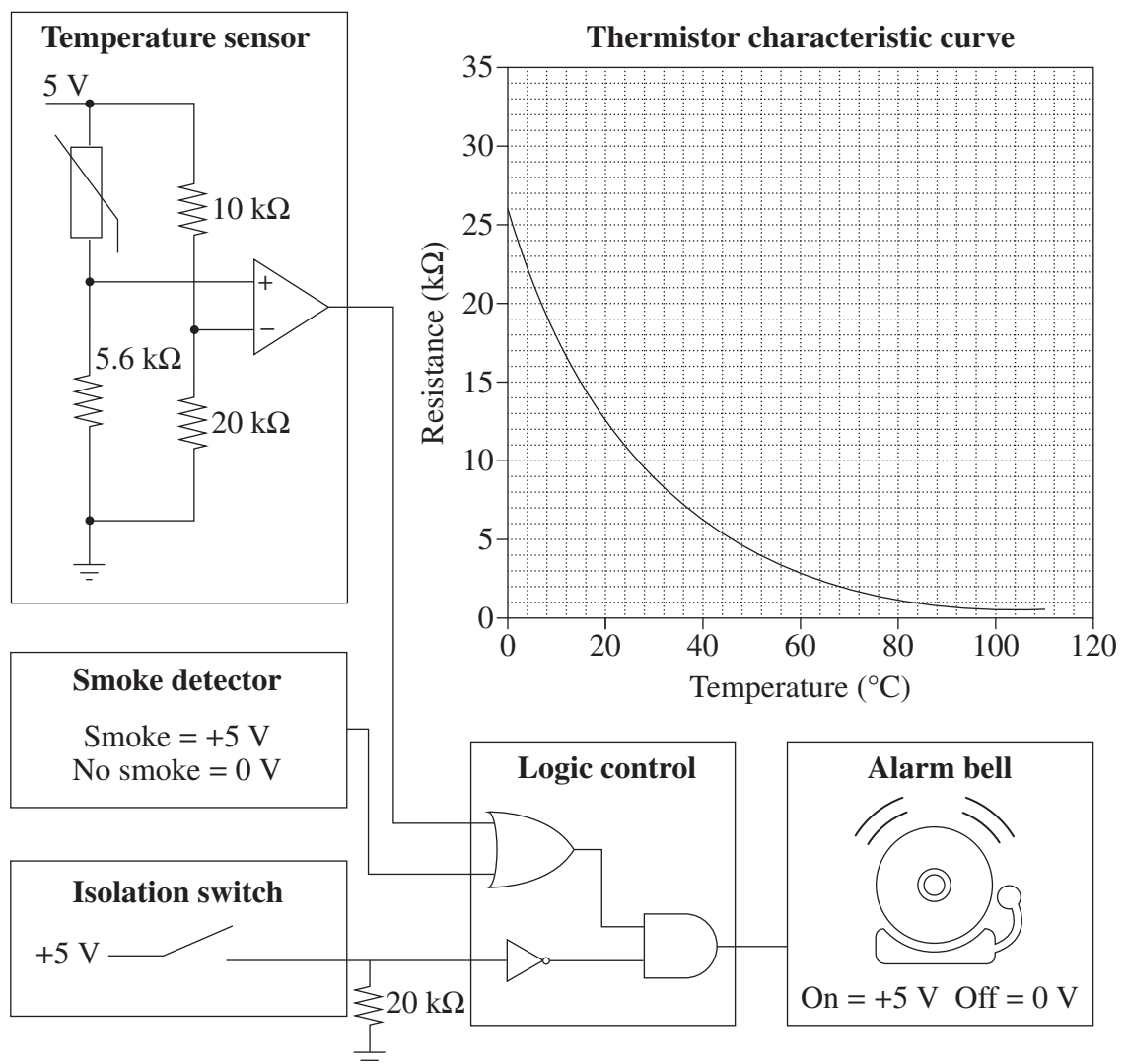
Relate information in the graph to the evolution of computer technology.

Question 35 continues on page 43

Question 35 (continued)

Answer parts (d) and (e) of the question on pages 6–8 of the Section II Writing Booklet. Start each part of the question on a new page.

- (d) Explain how an electronic circuit can use a relay to switch high-power devices. 4
In your answer, refer to the roles of the components of the relay.
- (e) The diagrams show a circuit and the characteristic curve of the thermistor used. 7



Explain the operation of this circuit. Support your answer with a truth table and data from the thermistor characteristic curve.

End of paper

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Physics

DATA SHEET

Charge on electron, q_e	$-1.602 \times 10^{-19} \text{ C}$
Mass of electron, m_e	$9.109 \times 10^{-31} \text{ kg}$
Mass of neutron, m_n	$1.675 \times 10^{-27} \text{ kg}$
Mass of proton, m_p	$1.673 \times 10^{-27} \text{ kg}$
Speed of sound in air	340 m s^{-1}
Earth's gravitational acceleration, g	9.8 m s^{-2}
Speed of light, c	$3.00 \times 10^8 \text{ m s}^{-1}$
Magnetic force constant, $\left(k \equiv \frac{\mu_0}{2\pi}\right)$	$2.0 \times 10^{-7} \text{ N A}^{-2}$
Universal gravitational constant, G	$6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Mass of Earth	$6.0 \times 10^{24} \text{ kg}$
Planck constant, h	$6.626 \times 10^{-34} \text{ J s}$
Rydberg constant, R (hydrogen)	$1.097 \times 10^7 \text{ m}^{-1}$
Atomic mass unit, u	$1.661 \times 10^{-27} \text{ kg}$ $931.5 \text{ MeV}/c^2$
1 eV	$1.602 \times 10^{-19} \text{ J}$
Density of water, ρ	$1.00 \times 10^3 \text{ kg m}^{-3}$
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

FORMULAE SHEET

$$v = f\lambda$$

$$I \propto \frac{1}{d^2}$$

$$\frac{v_1}{v_2} = \frac{\sin i}{\sin r}$$

$$E = \frac{F}{q}$$

$$R = \frac{V}{I}$$

$$P = VI$$

$$\text{Energy} = VIt$$

$$v_{\text{av}} = \frac{\Delta r}{\Delta t}$$

$$a_{\text{av}} = \frac{\Delta v}{\Delta t} \quad \text{therefore} \quad a_{\text{av}} = \frac{v - u}{t}$$

$$\Sigma F = ma$$

$$F = \frac{mv^2}{r}$$

$$E_k = \frac{1}{2}mv^2$$

$$W = Fs$$

$$p = mv$$

$$\text{Impulse} = Ft$$

$$E_p = -G \frac{m_1 m_2}{r}$$

$$F = mg$$

$$v_x^2 = u_x^2$$

$$v = u + at$$

$$v_y^2 = u_y^2 + 2a_y \Delta y$$

$$\Delta x = u_x t$$

$$\Delta y = u_y t + \frac{1}{2}a_y t^2$$

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$$

$$F = \frac{Gm_1 m_2}{d^2}$$

$$E = mc^2$$

$$l_v = l_0 \sqrt{1 - \frac{v^2}{c^2}}$$

$$t_v = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$m_v = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

FORMULAE SHEET

$$\frac{F}{l} = k \frac{I_1 I_2}{d}$$

$$d = \frac{1}{p}$$

$$F = BIl \sin \theta$$

$$M = m - 5 \log_{10} \left(\frac{d}{10} \right)$$

$$\tau = Fd$$

$$\frac{I_A}{I_B} = 100^{(m_B - m_A)/5}$$

$$\tau = nBIA \cos \theta$$

$$m_1 + m_2 = \frac{4\pi^2 r^3}{GT^2}$$

$$\frac{V_p}{V_s} = \frac{n_p}{n_s}$$

$$F = qvB \sin \theta$$

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

$$E = \frac{V}{d}$$

$$\lambda = \frac{h}{mv}$$

$$E = hf$$

$$c = f\lambda$$

$$A_0 = \frac{V_{\text{out}}}{V_{\text{in}}}$$

$$Z = \rho v$$

$$\frac{V_{\text{out}}}{V_{\text{in}}} = -\frac{R_{\text{f}}}{R_{\text{i}}}$$

$$\frac{I_r}{I_0} = \frac{[Z_2 - Z_1]^2}{[Z_2 + Z_1]^2}$$

PERIODIC TABLE OF THE ELEMENTS

1 H 1.008 Hydrogen		KEY										2 He 4.003 Helium					
3 Li 6.941 Lithium		4 Be 9.012 Beryllium		Atomic Number Symbol		79 Au 197.0 Gold		Standard Atomic Weight Name		5 B 10.81 Boron		6 C 12.01 Carbon	7 N 14.01 Nitrogen	8 O 16.00 Oxygen	9 F 19.00 Fluorine	10 Ne 20.18 Neon	
										13 Al 26.98 Aluminium	14 Si 28.09 Silicon	15 P 30.97 Phosphorus	16 S 32.07 Sulfur	17 Cl 35.45 Chlorine	18 Ar 39.95 Argon		
19 K 39.10 Potassium	20 Ca 40.08 Calcium	21 Sc 44.96 Scandium	22 Ti 47.87 Titanium	23 V 50.94 Vanadium	24 Cr 52.00 Chromium	25 Mn 54.94 Manganese	26 Fe 55.85 Iron	27 Co 58.93 Cobalt	28 Ni 58.69 Nickel	29 Cu 63.55 Copper	30 Zn 65.38 Zinc	31 Ga 69.72 Gallium	32 Ge 72.64 Germanium	33 As 74.92 Arsenic	34 Se 78.96 Selenium	35 Br 79.90 Bromine	36 Kr 83.80 Krypton
37 Rb 85.47 Rubidium	38 Sr 87.61 Strontium	39 Y 88.91 Yttrium	40 Zr 91.22 Zirconium	41 Nb 92.91 Niobium	42 Mo 95.96 Molybdenum	43 Tc Technetium	44 Ru 101.1 Ruthenium	45 Rh 102.9 Rhodium	46 Pd 106.4 Palladium	47 Ag 107.9 Silver	48 Cd 112.4 Cadmium	49 In 114.8 Indium	50 Sn 118.7 Tin	51 Sb 121.8 Antimony	52 Te 127.6 Tellurium	53 I 126.9 Iodine	54 Xe 131.3 Xenon
55 Cs 132.9 Caesium	56 Ba 137.3 Barium	57–71 Lanthanoids	72 Hf 178.5 Hafnium	73 Ta 180.9 Tantalum	74 W 183.9 Tungsten	75 Re 186.2 Rhenium	76 Os 190.2 Osmium	77 Ir 192.2 Iridium	78 Pt 195.1 Platinum	79 Au 197.0 Gold	80 Hg 200.6 Mercury	81 Tl 204.4 Thallium	82 Pb 207.2 Lead	83 Bi 209.0 Bismuth	84 Po Astatine	85 At Radon	86 Rn Radon
87 Fr Francium	88 Ra Radium	Actinoids	Rutherfordium	Dubnium	Seaborgium	Bohrium	Hassium	Meitnerium	Darmstadtium	Roentgenium	Copernicium	Nihonium	Flerovium	Moscovium	Livermorium	Tennessee	Oganesson

Lanthanoids

57 La 138.9 Lanthanum	58 Ce 140.1 Cerium	59 Pr 140.9 Praseodymium	60 Nd 144.2 Neodymium	61 Pm Promethium	62 Sm 150.4 Samarium	63 Eu 152.0 Europium	64 Gd 157.3 Gadolinium	65 Tb 158.9 Terbium	66 Dy 162.5 Dysprosium	67 Ho 164.9 Holmium	68 Er 167.3 Erbium	69 Tm 168.9 Thulium	70 Yb 173.1 Ytterbium	71 Lu 175.0 Lutetium
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Actinoids

89 Ac Actinium	90 Th 232.0 Thorium	91 Pa 231.0 Protactinium	92 U 238.0 Uranium	93 Np Neptunium	94 Pu Plutonium	95 Am Americium	96 Cm Curium	97 Bk Berkelium	98 Cf Californium	99 Es Einsteinium	100 Fm Fermium	101 Md Mendelevium	102 No Nobelium	103 Lr Lawrencium
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Standard atomic weights are abridged to four significant figures.

Elements with no reported values in the table have no stable nuclides.

Information on elements with atomic numbers 113 and above is sourced from the International Union of Pure and Applied Chemistry Periodic Table of the Elements (November 2016 version).
The International Union of Pure and Applied Chemistry Periodic Table of the Elements (February 2010 version) is the principal source of all other data. Some data may have been modified.

2018 HSC Physics Marking Guidelines

Section I, Part A

Multiple-choice Answer Key

Question	Answer
1	A
2	B
3	A
4	D
5	B
6	B
7	A
8	D
9	A
10	C
11	D
12	A
13	B
14	C
15	D
16	A
17	C
18	B
19	B
20	C

Section I, Part B

Question 21 (a)

Criteria	Marks
• Identifies both equal force AND opposite directions	2
• Provides some relevant information	1

Sample answer:

The force of Earth on the moon is equal in magnitude to the force of the moon on Earth, but in the opposite direction.

Question 21 (b)

Criteria	Marks
• Correctly compares the mass and weight of the person on the moon and on Earth quantitatively	2
• Provides some relevant information	1

Sample answer:

Mass does not change due to gravity so the mass will be 70 kg on Earth and on the moon.

$$W = mg$$

$$W_m = 70 \times 1.6 = 112$$

Weight on moon = 112 N

$$W_E = 70 \times 9.8 = 686$$

Weight on Earth is 686 N

Question 22 (a)

Criteria	Marks
• Relates the movement of the disc to the effect of the rotating magnet	3
• Outlines the movement of the disc and/or the effect of the magnet	2
• Provides some relevant information	1

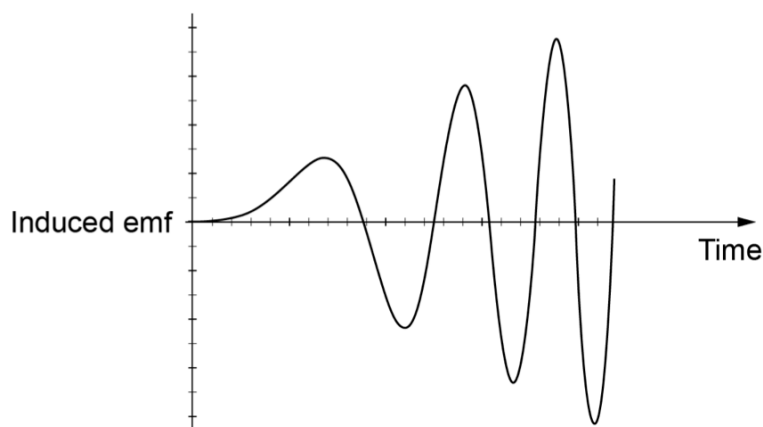
Sample answer:

As the magnet spins there is a change in magnetic flux in the metal disc, inducing eddy currents which, by Lenz's law and conservation of energy, will tend to create magnetic fields that will oppose the change. This will have the effect of the disc trying to keep up with the magnet and it will therefore spin in the same direction.

Question 22 (b)

Criteria	Marks
• Draws a graph showing all correct features	3
• Draws a graph showing some correct features	2
• Draws a graph with a correct feature	1

Sample answer:



[The graph should show a curve that is increasing in both amplitude and frequency.]

Question 23 (a)

Criteria	Marks
• Outlines energy changes in the electron gun and the screen	2
• Identifies an energy change	1

Sample answer:

The electron gains kinetic energy as it passes through the electron gun. At the screen the electron's kinetic energy is converted to light energy.

Question 23 (b)

Criteria	Marks
• Relates features of the components to forming an image on the screen	3
• Outlines some features of the components	2
• Provides some relevant information	1

Sample answer:

The components labelled Y provide a force on moving electrons, causing them to deflect. By varying the strength of the magnetic field, the force on the electron can be controlled to steer the electron beam to different parts of the screen.

Question 24 (a)

Criteria	Marks
• Outlines TWO safety features	2
• Provides a safety feature	1

Sample answer:

Earth line above the high-voltage transmission lines to protect them from lightning strikes and insulating discs used to attach transmission lines to power poles to prevent short-circuiting.

Question 24 (b)

Criteria	Marks
• Provides correct calculation of magnitude and direction of current	3
• Provides some steps to calculate the magnitude of the current OR • Substitutes into a relevant equation and shows direction of current	2
• Substitutes into a relevant equation OR • Shows direction of current	1

Sample answer:

$$I_1 = 20 \text{ A}, \quad I_2 = 20 \text{ A}, \quad I_3 = ?$$

$$d_1 = 0.3, \quad d_2 = 0.75$$

$$\frac{F_1}{I} = \frac{F_2}{I}, \text{ therefore } \frac{kI_1I_2}{d_1} = \frac{kI_2I_3}{d_2}$$

$$\text{So } \frac{20 \times 20}{0.3} = \frac{20I_3}{0.75}, \text{ thus } I_3 = 50 \text{ A travelling in the same direction as Y.}$$

Question 25 (a)

Criteria	Marks
• States ONE benefit and ONE limitation	2
• States a benefit or a limitation	1

Sample answer:

Benefit – no energy loss in transmission

Limitation – energy required to keep the transmission lines cold

Question 25 (b)

Criteria	Marks
• Contrasts features of electrical conduction in the two types of materials	4
• Outlines features of electrical conduction in one material and identifies a feature of conduction in the other material	3
• Outlines a feature of electrical conduction in either material OR • Identifies a feature of electrical conduction in each material	2
• Provides some relevant information	1

Sample answer:

In a room temperature metal, electrons move independently of each other, whereas in a superconductor electrons move in pairs.

In a room temperature metal, interaction between electrons moving through a lattice and the thermal lattice vibrations gives rise to resistance, whereas in a superconductor the electrons move without resistance through the crystal lattice which is distorted only by the electron pairs.

Question 26

Criteria	Marks
• Outlines similarities and differences between the effects of electric and gravitational fields • Refers to definitions of the fields	4
• Outlines similarities and/or differences between the effects of electric and gravitational fields	3
• Shows some understanding of the effects of electric fields and/or gravitational fields	2
• Provides some relevant information	1

Sample answer:

An electric field is a region in space where a force acts on a charged particle. A gravitational field is a region in space where an object with mass affects other objects with mass.

Electric field strength is given by $E = F/q$ (force per unit charge), whereas gravitational field strength is given by $g = F/m$ (force per unit mass).

Both fields produce forces of attraction, but only electric fields repel (due to like charges). Electric and gravitational fields vary in strength due to magnitudes of the charge or mass respectively.

Both fields decrease in strength with the square of the separation distance from a point charge or mass.

Answers could include

Gravity is a very weak fundamental force compared to the forces dependent on charge.

Question 27 (a)

Criteria	Marks
- Identifies an error - Describes the effect	3
- Identifies an error - Identifies the effect	2
Identifies an error or and effect	1

Sample answer:

Error: The camera is off centre

Effect: When the object is closer to the camera, it will appear to be moving faster than when it is further away, therefore it will give an incorrect measurement of velocity.

Answers could include:

Error: Ruler closer to the camera than the trajectory

Effect: Distance calculated on the trajectory using the ruler will be greater than the true distance and hence the calculated velocity will be greater than the true value.

Others: Camera too close/non-linearity of distances etc; ruler off centre/non-linearity of distances etc.

Question 27 (b)

Criteria	Marks
Describes quantitatively and qualitatively the velocity and acceleration related to the graphs	3
Describes the velocity and acceleration related to the graphs	2
- Identifies features of the motion related to the graph(s) OR - Describes the velocity or acceleration related to the graph(s)	1

Sample answer:

The first graph shows that the ball is moving horizontally at a constant speed of 1.6 m s^{-1} . The second graph shows that the ball is accelerating at a constant rate in the vertically downward direction.

Answers could include:

$$\text{velocity} = \frac{-1.1 - (-0.3)}{0.5} = \frac{-0.8}{0.5} = -1.6 \text{ m s}^{-1}$$

$$\text{acceleration} = \frac{-2.4 - 2.4}{0.5} = -9.6 \text{ m s}^{-2}$$

Question 28 (a)

Criteria	Marks
• Shows that the change in potential energy is $1.26 \times 10^7 \text{ J}$	2
• Substitutes some information into a relevant equation	1

Sample answer:

Initial potential energy of mass

$$E_{\text{pi}} = -\frac{GMm}{r} = \frac{-6.67 \times 10^{-11} \times 7.35 \times 10^{22} \times 20}{1.74 \times 10^6}$$

$$= -5.64 \times 10^7 \text{ J}$$

Final potential energy of mass

$$E_{\text{pf}} = \frac{-6.67 \times 10^{-11} \times 7.35 \times 10^{22} \times 20}{2.24 \times 10^6}$$

$$= -4.38 \times 10^7 \text{ J}$$

$$\Delta E_{\text{p}} = E_{\text{pf}} - E_{\text{pi}} = 1.26 \times 10^7 \text{ J}$$

Question 28 (b)

Criteria	Marks
• Applies a correct process to calculate the velocity	3
• Applies the law of conservation of energy	2
• Substitutes some information into a relevant equation	1

Sample answer:

Initial $E_{\text{ki}} = \frac{1}{2}mu^2 = \frac{1}{2} \times 20 \times 1200^2$

$$= 1.44 \times 10^7 \text{ J}$$

Final $E_{\text{kf}} = E_{\text{ki}} - \Delta E_{\text{p}}$

$$= 1.44 \times 10^7 - 1.26 \times 10^7$$

$$= 1.8 \times 10^6 \text{ J}$$

$$E_{\text{kf}} = \frac{1}{2}mv^2$$

$$v = \sqrt{\frac{2E_{\text{kf}}}{M}} = \sqrt{\frac{2 \times 1.8 \times 10^6}{20}}$$

$$= 424 \text{ m s}^{-1}$$

Question 29 (a)

Criteria	Marks
• Shows how light contributes to the flow of current in each of the cells	4
• Shows how light contributes to the flow of current in one of the cells and links light to the flow of current in the other cell	3
• Links light to the flow of current in each of the cells OR • Shows how light contributes to the flow of current in one of the cells	2
• Provides some relevant information	1

Sample answer:

In a vacuum photocell, incoming photons strike a photocathode *A*, knocking electrons out of its surface, which are attracted to an anode *B*, due to the potential difference applied by the battery, thus producing a current.

In a solar cell, a potential difference is set up by the *p* and *n* junction. When incoming photons of light hit the electrons the *n*-type semiconductor electrons are pushed through the external circuit, thus creating a current, which is electricity for the load.

Question 29 (b)

Criteria	Marks
• Explains how one advance in scientific understanding and one advance in technology have led to the use of semiconductors	4
• Describes an advance in either understanding or technology that have led to the use of semiconductors AND • Identifies an advance in the other	3
• Shows some understanding of advance in scientific understanding and/or technology in relation to the use of semiconductors	2
• Provides some relevant information	1

Sample answer:

Our increased understanding of the band structure model has led to the doping of materials. Advances in technology have allowed silicon to be easily purified to make an intrinsic semiconductor ready for doping. Doping is used to produce different types of semiconductors: *n*-type, and *p*-type. Layering these together allows us to make diodes and transistors which have many uses.

Answers could include:

<i>Scientific understanding</i>	Band structure	Effects of doping
<i>Technology</i>	Purification	Fabrication

Question 30

Criteria	Marks
- Explains the effects of the increased demand on the components of the system - Refers to voltage, current and energy	6
- Explains the effects of the increased demand on components of the system - Refers to voltage, current and/or energy	5
- Describes an effect of the increased demand on a component of the system - Makes reference to voltage, current OR energy	4
- Outlines an effect of the increased demand on the system OR - Outlines how components of the system work together to transmit power	3
- Identifies effects of the increased demand on the system OR - Outlines the operation of a component of the system	2
Provides some relevant information	1

Sample answer:

The increased demand for energy can be analysed in terms of the effect on voltages, current and energy losses.

Voltages: The output voltage from the power station, the transmission line voltage and the supply voltage to the houses remains approximately constant.

Current: An increase in the total current being drawn from T2 by the houses produces an increase in the transmission line current and the output current from the power station. If the former doubles, then the transmission line current and the current output of the power station approximately double.

Energy losses: Heat is produced in the transmission line wires and other wires due to their resistance. As the current increases the heat produced in the wires increases. Heat is also produced by eddy currents in the transformer cores, and this increases as the current increases.

Because of the resistance of the wires, relatively small voltage drops occur along these conductors and these increase as more appliances are turned on in the houses. This results in slight decreases in the voltage inputs and outputs at each of the transformers.

Answers could include:

Reference to parallel circuits in the houses

Section II

Question 31 (a) (i)

Criteria	Marks
• Provides TWO methods	2
• Provides ONE method	1

Sample answer:

Radiometric and gravitational methods

Answers could include:

Magnetic, palaeomagnetic or geothermal

Question 31 (a) (ii)

Criteria	Marks
• Shows a sound understanding of how a suitable geophysical method provides information about Earth's physical properties	3
• Outlines how a suitable geophysical method provides information about Earth's physical properties	2
• Provides some relevant information	1

Sample answer:

Radiometric methods are used for dating specimens. The radioactive decay of elements in a rock or mineral is measured and compared to determine its age. By using radioactive elements with differing half-lives, ages can be determined for specimens from a few thousand years to billions of years of age.

Answers could include:

Other answers could describe information about:

Magnetic, gravitational, palaeomagnetic or geothermal methods

Question 31 (b) (i)

Criteria	Marks
• Explains a relevant feature	2
• Provides some relevant information	1

Sample answer:

The altitude of NOAA-20 is much less than that of geostationary satellites, hence photographs will provide more detail about storms and cloud systems.

Answers could include:

Due to NOAA-20's short period polar orbit, it regularly passes over the entire surface of Earth to photograph changes in the atmosphere.

Question 31 (b) (ii)

Criteria	Marks
• Applies a correct process to calculate the mass of Earth	3
• Shows some appropriate calculations	2
• Provides some relevant information	1

Sample answer:

$$\begin{aligned}
 \frac{r^3}{T^2} &= \frac{GM}{4\pi^2} \\
 M &= \frac{r^3 4\pi^2}{GT^2} \\
 &= (r_E + \text{altitude})^3 \times \frac{4\pi^2}{GT} \\
 &= \frac{(6.37 \times 10^6 + 8.7 \times 10^5)^3 \times 39.48}{6.67 \times 10^{-11} \times (100 \times 60)^2} \\
 &= 6.24 \times 10^{24}
 \end{aligned}$$

The mass of Earth is calculated to be 6.24×10^{24} kg

Question 31 (c)

Criteria	Marks
- Shows an understanding of both Richer's and Newton's use of data obtained from experiments involving pendulums - Explains how both independently used the data to determine the shape of Earth	4
- Shows an understanding of both Richer's and Newton's use of data obtained from experiments involving pendulums - Explains how one of them used the data to determine the shape of Earth	3
Shows some understanding of Richer's and/or Newton's use of data obtained from experiments involving pendulums	2
Provides some relevant information	1

Sample answer:

Richer found that the rate of swing of a pendulum at the Equator was slightly slower than the rate of swing of a similar pendulum in Paris. Since the rate of swing of a pendulum is directly related to the acceleration due to gravity, this suggested that the acceleration due to gravity was less at the Equator than in Paris. Newton used this information to explain that the Earth was flattened at the poles and bulged at the Equator.

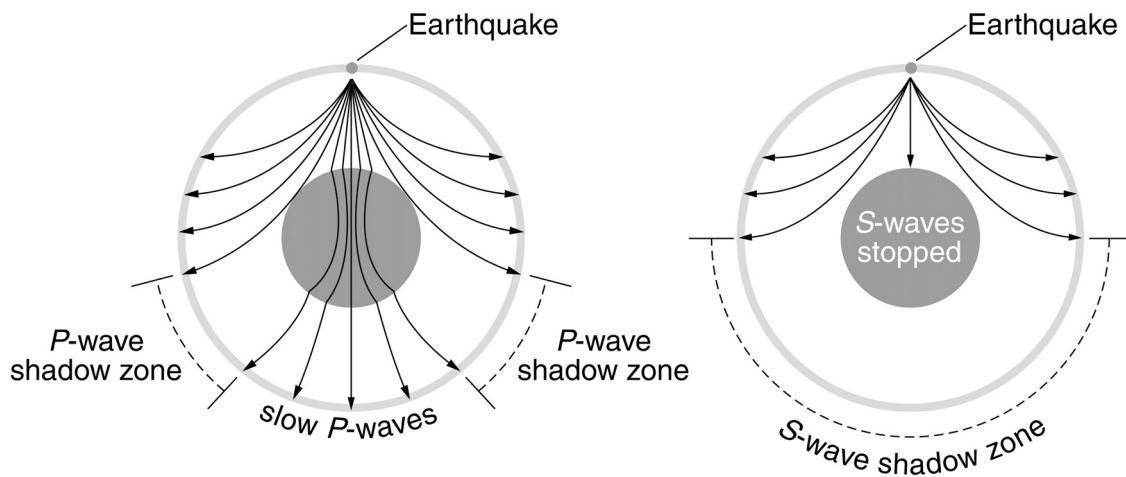
Question 31 (d)

Criteria	Marks
- Shows how seismic data are used to provide a model of Earth's core - Supports answer with a diagram	4
Shows how seismic data are used to provide a model of Earth's core	3
Shows some understanding of Earth's core and/or the use of seismic data	2
Provides some relevant information	1

Sample answer:

Seismic waves may reflect (bounce) or refract (bend and change speed) depending on the materials they pass through. *P*-waves (compression waves) and *S*-waves (shear waves) both travel through solids, but *S*-waves cannot travel through liquids.

Seismic waves can be detected around Earth's surface, as shown in the diagrams.



Scientists have used this seismic data to produce a model of Earth's core: a liquid outer core around a solid inner core.

Question 31 (e)

Criteria	Marks
• Explains the role of palaeomagnetic data in providing evidence that supports the theory of plate tectonics	7
• Describes the role of palaeomagnetic data in relation to the theory of plate tectonics	6
• Outlines the role of palaeomagnetic data in relation to the theory of plate tectonics	4–5
• Shows some understanding of palaeomagnetic data and/or the theory of plate tectonics	2–3
• Provides some relevant information	1

Sample answer:

Certain minerals in rocks lock in a record of the direction and intensity of Earth's magnetic field when they form. Palaeomagnetism is the study of this record in ancient rocks.

Mirror image patterns of magnetic reversal in the sea floor on either side of mid-ocean ridges provided strong evidence for sea-floor spreading when correlated with data from which the ages of the rocks can be determined.

Apparent polar wander paths determined from palaeomagnetic data provided evidence for continental drift.

Together, continental drift and sea-floor spreading formed the basis of the theory of plate tectonics which describes the Earth's surface as a series of major and minor plates which move around the surface of the Earth due to convection currents in the mantle.

Answers may include:

Use of such data to explain earthquake and volcanic activity, mountain building and oceanic trench formation.

Question 32 (a) (i)

Criteria	Marks
• Identifies two specific uses of MRI	2
• Identifies one specific use of MRI OR • Outlines a general use of MRI	1

Sample answer:

MRI is used to detect cancerous tissues and to distinguish between grey and white matter in the brain.

Question 32 (a) (ii)

Criteria	Marks
• Describes effects of the external field on the protons	3
• Identifies effects of the external field on the protons OR • Describes an effect of the external field on the protons	2
• Identifies an effect of the external field on the protons	1

Sample answer:

Hydrogen nuclei in water molecules tend to align their spins with the field. The axes of rotation are not exactly parallel to the external field direction and the spins of these protons precess around the direction of the applied external field at a rate that is dependent on the strength of the field. The hydrogen nuclei align parallel or antiparallel to the field with a small minority in the parallel direction.

Question 32 (b) (i)

Criteria	Marks
• Shows how the behaviour of waves <i>A</i> and <i>B</i> are similar and/or different	2
• Shows some understanding of the behaviour of waves <i>A</i> or <i>B</i>	1

Sample answer:

The proportion of the ultrasound that is transmitted (or reflected) at the boundary is the same for both *A* and *B*.

Both waves have some of their energy reflected at the boundary and some of their energy transferred across the boundary.

Question 32 (b) (ii)

Criteria	Marks
• Applies a correct process to calculate the acoustic impedance of fat tissue	3
• Provides some main steps to calculate the acoustic impedance of fat tissue	2
• Substitutes some information into a relevant formula	1

Sample answer:

$$Z_{\text{kidney}} = \rho v = 1050 \times 1560 = 1.64 \times 10^6 \text{ Rayls}$$

$$\frac{I_r}{I_o} = \frac{(Z_2 - Z_1)^2}{(Z_2 + Z_1)^2}$$

$$0.01 = \frac{(1.64 \times 10^6 - Z_1)^2}{(1.64 \times 10^6 + Z_1)^2}$$

$$0.1 = \frac{1.64 \times 10^6 - Z_1}{1.64 \times 10^6 + Z_1}$$

$$0.1 \times 1.64 \times 10^6 + 0.1 \times Z_1 = 1.64 \times 10^6 - Z_1$$

$$1.1 \times Z_1 = 1.64 \times 10^6 - 1.64 \times 10^5 = 1.476 \times 10^6$$

$$\text{Hence } Z_1 = 1.34 \times 10^6 \text{ Rayls}$$

Question 32 (c)

Criteria	Marks
- Justifies the use of I-124 as a PET medical diagnostic tool - Supports answer quantitatively	4
- Outlines reason(s) for using I-124 as a medical diagnostic tool - Links its use quantitatively to its half-life	3
Shows some understanding of the use of I-124 as a medical diagnostic tool	2
Provides some relevant information	1

Sample answer:

Iodine-124 is a positron emitter and hence could be used for PET scans. Each positron emitted by this isotope in the body undergoes annihilation with an electron, producing two gamma photons which can be detected outside the body. Gamma rays are not strongly ionising, making them safe for diagnostic use.

The gamma radiation can be detected once it has left the body, providing information about the parts of the body where it was produced.

The half-life of approximately 100 hours makes it suitable for use in the body because the potentially damaging radiation does not persist in the body long enough to be dangerous.

Question 32 (d)

Criteria	Marks
Explains how the Doppler effect is used for medical imaging	4
Describes the use of the Doppler effect in medical imaging	3
- Outlines features of the Doppler effect OR - Outlines features of ultrasound imaging	2
Provides some relevant information	1

Sample answer:

When ultrasound waves are reflected from a moving surface, the frequency of the reflected wave is higher if the surface is moving towards the incident wave and lower if it is moving away. The greater the velocity of the reflecting surface, the greater the change in frequency. This is the Doppler effect.

Information about the angle between the directions of the velocities of the ultrasound and the reflecting surface can be determined to produce more accurate images.

The position of the reflecting tissue can be calculated knowing the velocity of the ultrasound in the tissues and the time for the reflected signal to be detected.

Doppler ultrasound is therefore used to analyse movement of tissues such as heart valves and the flow of blood.

Question 32 (e)

Criteria	Marks
- Compares the production of the longitudinal CAT image with an X-ray image of the same area - Compares the benefits of the CAT image provided with an X-ray image of the same area	7
Describes the production and benefits of CAT and X-ray images including some comparisons	6
Outlines production and benefits of CAT and X-ray images	4–5
Shows some understanding of CAT images and/or X-ray images	2–3
Provides some relevant information	1

Sample answer:

The production of both types of scans is dependent on differences in the absorption of X-rays passed through the body by different types and depths of tissue. Conventional X-rays are cheaper.

A conventional X-ray image of the neck is made by passing X-rays in a wide beam transversely through the neck producing a longitudinal image of the neck in a single plane, which is the result of the absorption of the X-rays by all the intervening tissues. The image can be ambiguous.

In contrast to this, a CAT scan is produced by a narrow beam of X-rays that are passed through the person from a source that rotates in the transverse plane around the person's body, and which are detected by a sensor that also moves so that it is always opposite the X-ray source. This data is processed by a computer to produce an image that is a cross-section of the body. The benefit is that the CAT image is clearer and less ambiguous than the conventional X-ray image. The CAT scan also produces better soft tissue differentiation.

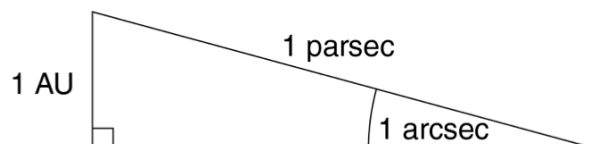
In the case of the CAT scan shown, it shows structures in the longitudinal plane, as would be the case with a conventional X-ray. This is achieved by combining multiple transverse images allowing computer software to produce an image in a single plane through the body at any angle.

Question 33 (a) (i)

Criteria	Marks
• Uses a diagram to define the parsec	2
• Provides some relevant information	1

Sample answer:

One parsec is the distance from Earth to a point that has an annual parallax of one arcsec.



Question 33 (a) (ii)

Criteria	Marks
• Outlines TWO limitations	3
• Identifies TWO limitations OR	2
• Outlines ONE limitation	
• Provides some relevant information	1

Sample answer:

As Earth's atmosphere causes distortion of incoming light, light waves are refracted by the gas particles and as a result resolution of ground-based telescopes can be low. Only large angles of parallax can be measured accurately, which means only very small distances in space can be measured.

Question 33 (b) (i)

Criteria	Marks
• Contrasts the reasons for the variation in brightness of these two stars	2
• Provides some relevant information	1

Sample answer:

Eclipsing binary stars' brightness varies due to the relative motion between the two stars whereas Cepheid's varying brightness is due to the star's core activity.

Question 33 (b) (ii)

Criteria	Marks
• Applies correct procedure to calculate distance	3
• Provides some main steps to calculate distance	2
• Substitutes some information into a relevant formula	1

Sample answer:

From graph, period = 6 days, therefore $M = -3.2$ and $m = 16.3$.

$$M = m - 5 \log_{10} \left(\frac{d}{10} \right)$$

$$\log_{10} \left(\frac{d}{10} \right) = \frac{16.3 - (-3.2)}{5}$$

$$\log_{10} \left(\frac{d}{10} \right) = 3.9$$

$$d = 10^{3.9} \times 10$$

$$d = 79\,432.8 = 79 \text{ kpc}$$

Question 33 (c)

Criteria	Marks
Describes physical and chemical changes through the remaining stages of the star's life	4
- Identifies some physical and chemical changes through the remaining stages of the star's life - Describes at least one of the changes	3
Shows some understanding of how star X evolves	2
Provides some relevant information	1

Sample answer:

The next stage of its evolutionary path would be a red supergiant where its size will increase dramatically but the luminosity would stay the same. Helium burning starts in the core and changes to fusion between helium and carbon to form oxygen. As the temperature rises carbon fuses to form heavier elements and iron which causes the core to collapse forming a supernova explosion which results in an increase in its luminosity leading to a gravitational collapse into a black hole.

Question 33 (d)

Criteria	Marks
- Describes differences between photoelectric and photographic methods - Relates differences to the increased understanding provided	4
- Describes differences between photoelectric and photographic methods OR - Describes a difference between photoelectric and photographic methods and relates the difference to the increased understanding provided	3
Shows some understanding of photographic and/or photoelectric methods	2
Provides some relevant information	1

Sample answer:

With photographic methods each image contains many stars whereas photoelectric technologies using charged-coupled devices (CCD) can isolate the light from a single star and thoroughly analyse it.

CCD images show fainter objects, collect more effectively across the entire spectrum (including infrared) and can be collected and processed more rapidly than photographic methods. Due to its digitisation, computers can be used to enhance, enlarge, add or subtract selected colours to assist in enhancing specific features.

This gives a greater understanding of the star in terms of its surface temperature, chemical composition, mass, density, rotational and translational velocity.

Question 33 (e)

Criteria	Marks
- Explains how the three spectra are used to analyse surface temperature and chemical composition of stars - Refers to observations that can be made in a school laboratory	7
- Explains how at least two of the spectra are used to analyse surface temperature and chemical composition of stars with reference to observations that can be made in a school laboratory OR - Explains how the three spectra are used to analyse surface temperature and/or chemical composition of stars with reference to observations that can be made in a school laboratory	6
- Outlines how black body spectra and/or emission spectra and/or absorption spectra are used to analyse surface temperature and/or chemical composition of stars - Better responses should refer to observations that can be made in a school laboratory	4–5
- Shows some understanding of the use of black body spectra and/or emission spectra and/or absorption spectra to analyse stars AND/OR - Refers to observations that can be made in a school laboratory	2–3
Provides some relevant information	1

Sample answer:

A black body produces a radiation curve that is dependent on the temperature of the body. A star acts as a black body and the characteristic peak of the curve reflects the temperature of the surface of the star.

A black body produces a continuous spectrum, so the spectrum of stars is continuous. Absorption lines are seen in the spectrum due to cooler gas particles in the atmosphere around the stars absorbing certain wavelengths of energy resulting in an absorption spectrum which identifies the atoms in the gas clouds.

Using different gas discharge tubes (hydrogen, helium, neon) and a spectroscope, unique spectra can be observed in a school laboratory. This can be used to compare with spectral lines produced by stars to identify their chemical composition. Incandescent bulbs can demonstrate a continuous spectrum curve.

Question 34 (a) (i)

Criteria	Marks
• Identifies TWO limitations of the Bohr model of the atom	2
• Identifies a limitation of the Bohr model of the atom	1

Sample answer:

- Only works for hydrogen
- Doesn't account for variation in spectral line intensity

Answers could include:

- Doesn't account for hyperfine spectral lines
- Doesn't account for the Zeeman effect.

Question 34 (a) (ii)

Criteria	Marks
• Shows how the Bohr model accounts for the emission spectrum of hydrogen	3
• Relates a feature of the Bohr model to the production of the emission spectrum	2
• Provides some relevant information	1

Sample answer:

The Bohr model consists of an electron occupying only very specific energy levels in orbiting the positive nucleus. According to this model, the emission spectrum of hydrogen is created when an excited electron falls from a higher energy level to a lower energy level with a corresponding release of a photon with energy a multiple of hf .

Answers could include:

Reference to a well-labelled diagram

Question 34 (b) (i)

Criteria	Marks
• Shows an understanding of how the law of conservation of momentum or energy was used to estimate the mass or energy of the particle	2
• Provides some relevant information	1

Sample answer:

When calculating the momentums of the three particles, alpha, X and proton, the law of conservation of momentum indicates that the mass of X is similar to the mass of a proton. This is how Chadwick discovered the neutron.

Question 34 (b) (ii)

Criteria	Marks
• Applies a correct procedure to calculate energy in joules	3
• Provides some main steps to calculate energy	2
• Substitutes some information into a relevant formula	1

Sample answer:

$$\text{Initial mass} = 4.0012 + 9.0122 = 13.0134u$$

$$\text{Final mass} = 12.000 + 1.0087 = 13.0087u$$

$$\text{Mass difference} = 13.0087 - 13.0134 = -0.0047u$$

$$\text{Mass in kg} = 0.0047 \times 1.661 \times 10^{-27} = 7.8067 \times 10^{-30} \text{ kg}$$

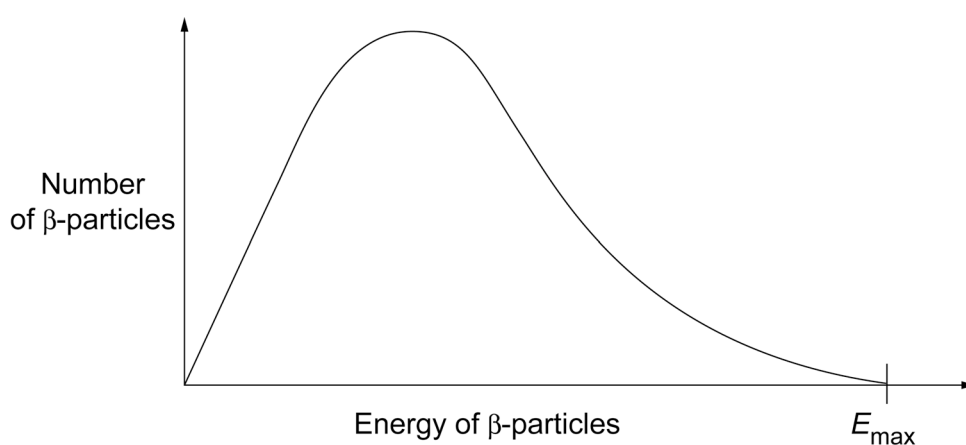
$$E = mc^2 = 7.8067 \times 10^{-30} \times (3.00 \times 10^8)^2 = 7.0 \times 10^{-13} \text{ J}$$

Question 34 (c)

Criteria	Marks
- Shows how Pauli inferred the existence and properties of a new particle - Supports answer with a graph	4
Shows an understanding of how Pauli inferred the existence and properties of a new particle	3
Shows some understanding of how Pauli inferred the existence and/or properties of a new particle	2
Provides some relevant information	1

Sample answer:

The graph below represents the measured energies of emitted beta particles from a nuclear reaction.



The law of conservation of energy states that any nuclear reaction would release a fixed amount of energy.

The graph shows β -particles are emitted with a range of energy implying that some other particle must exist that carries the difference in energy.

Pauli inferred a new particle, neutrino, which must be chargeless and have negligible mass.

Question 34 (d)

Criteria	Marks
• Describes changes in the nuclei of X and Y	4
• Identifies a change in the nucleus of X and nucleus of Y • Describes one of the changes	3
• Identifies a change in the nucleus of X and the nucleus of Y OR • Outlines a change in the nucleus of X or Y OR • Relates the trails to the emission of particles from X and Y	2
• Provides some relevant information	1

Sample answer:

Source X emits an alpha particle (He nucleus), hence atomic number decreases by 2 and mass decreases by 4.

Source Y emits a beta particle (electron) from the nucleus, hence mass change is insignificant, but atomic number increases by 1.

Question 34 (e)

Criteria	Marks
• Analyses the roles of both forces and particles using the standard model • Relates these to the current understanding of the atom	7
• Analyses the roles of forces and/or particles using the standard model • Relates these to the current understanding of the atom	6
• Describes the roles of forces and/or particles in relation to the standard model • Links these to the current understanding of the atom	4–5
• Shows some understanding of the standard model	2–3
• Provides some relevant information	1

Sample answer:

In the standard model, fundamental particles – quarks and leptons – interact through four fundamental forces. The constituents of protons and neutrons are three quarks. The strong force holds nucleons together in the nucleus, overcoming the electrostatic repulsion between protons. An electron is a lepton which has no constituents. The force of gravity attracts protons and neutrons within the nucleus but this force is negligible.

The electrostatic force results in a mutual repulsion of protons and an attraction for the electrons toward the nucleus. Electrons orbit the nucleus. The strong nuclear force counteracts the electrostatic force, but only within a limited range of subatomic distances. The weak force is involved in the process of radioactive decay.

Question 35 (a) (i)

Criteria	Marks
• Outlines TWO differences between an electric circuit and an electronic circuit	2
• Outlines a difference between an electric circuit and an electronic circuit	1

Sample answer:

Electric circuits are designed for transmitting power and therefore operate with higher voltages and currents than electronic circuits in which voltages and currents represent information or signals.

Question 35 (a) (ii)

Criteria	Marks
• Relates properties of silica to its use in electronic communication	3
• Relates a property of silica to its use in electronic communication	2
• Provides some relevant information	1

Sample answer:

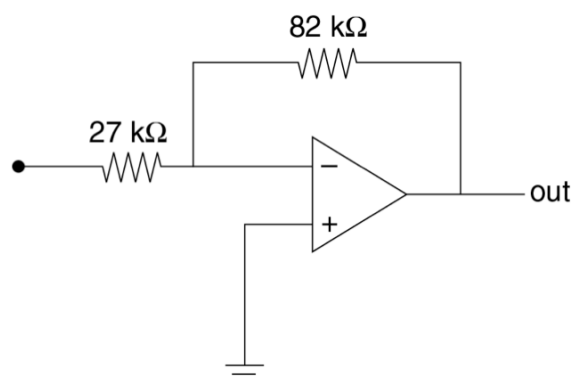
Silica can be easily drawn into fibres at high temperatures to produce optic fibres. Silica is optically transparent and can be made very pure, so that very little attenuation occurs over a long distance.

Silica has a refractive index of about 1.5, but it can vary such that the core of the fibre can be made to have a greater refractive index than the surrounding cladding, allowing total internal reflection to occur.

Question 35 (b) (i)

Criteria	Marks
• Draws the inverting amp with resistors connected in the correct place	2
• Provides some relevant information	1

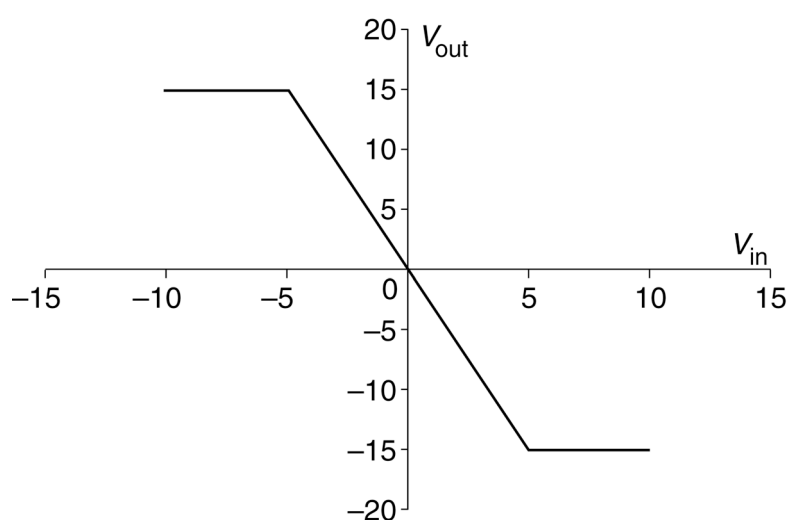
Sample answer:



Question 35 (b) (ii)

Criteria	Marks
• Draws a graph showing the correct features	3
• Draws a graph with some correct features	2
• Draws a graph with a correct feature OR calculates the correct gradient	1

Sample answer:



Question 35 (c)

Criteria	Marks
• Relates information in the graph to the evolution of computer technology • Refers to both the history and forward prediction of the graph	4
• Shows an understanding of the graph and the evolution of computer technology • Links the graph to the evolution of computer technology	3
• Shows some understanding of the graph and/or the evolution of computer technology	2
• Provides some relevant information	1

Sample answer:

Individual resistors have been replaced by integrated circuits containing many transistors. The graph shows that since 1970 the transistor count has increased at an exponential rate (by a factor of 10 roughly every 7–8 years). With this continued trend there has been an associated increase in computing power and portability of computers. Computers have gone from specialised devices in military, research and businesses to becoming ubiquitous devices in everyone's homes and in the palm of their hand. Today, small handheld devices have exponentially more computing power than early supercomputers that filled entire rooms. The graph also shows a predicted trend into the future. However, one has to be careful how far we extrapolate due to physical limits on transistor sizes, where components become so small that quantum effects become significant.

Question 35 (d)

Criteria	Marks
- Explains how an electronic circuit can use a relay to switch high-power devices - Refer to the roles of the components of the relay	4
Describes how an electronic circuit can use a relay to switch high-power devices	3
Shows some understanding of relays	2
Provides some relevant information	1

Sample answer:

An electronic circuit cannot handle high voltages or currents without damaging components. Relatively small power is required to operate an electromagnet which forms the input of the relay. This electromagnet pulls a lever which, via a pivot, pushes a switch contact closed. The switch contacts are electrically insulated from the lever in order to isolate the output from the input. Since it is a mechanical switch, it can be manufactured to handle much higher voltages and currents.

Answers may include:

A well-labelled diagram

Question 35 (e)

Criteria	Marks
- Explains the operation of the circuit - Supports explanation with a truth table and data from the thermistor characteristic curve	7
- Describes the operation of the circuit - Includes a truth table and/or data from the thermistor characteristic curve	6
- Outlines the operation of the circuit - Includes a truth table and/or data from the thermistor characteristic curve	4–5
Shows some understanding of the operation of the circuit and/or provides a truth table	2–3
Provides some relevant information	1

Sample answer:

For the temperature sensor to produce a high output, the input voltage must be greater than the reference voltage provided by the two fixed resistors. As the temp rises, R_t will decrease, increasing the input voltage.

$$\frac{R_t}{5.6} = \frac{10}{20}$$

$$R_t = 2.8 \text{ k}\Omega$$

On the graph this corresponds to 60°C

Inputs			Output
Smoke	Temp	Switch	
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

The alarm will be activated only if the switch is open and either or both the outputs (T_{out} , S_{out}) are high.

The alarm will be activated under the following conditions: The switch must be open and either the temp is above 60° or the smoke detector output is high.

2018 HSC Physics Mapping Grid

Section I Part A

Question	Marks	Content	Syllabus outcomes
1	1	9.2.3.2.1/3	H9
2	1	9.4.2.2.3	H2, H10
3	1	9.2.2.2.11	H6, H7
4	1	9.3.2.2.6	H7, H9
5	1	9.3.1.3.3	H9
6	1	9.3.2.3.3	H9
7	1	9.2.1.3.3/9.2.3.3.2	H9
8	1	9.4.3.2.2/3/7	H3, H7, H10
9	1	9.2.4.2.9/10	H6
10	1	9.3.1.2.3/5/6	H9
11	1	9.2.2.2.5/7	H9
12	1	9.4.1.3.3	H9
13	1	9.4.1.2.5, 9.4.1.2.8	H9
14	1	9.2.1.3.1/2	H9
15	1	9.4.4.2.2	H6
16	1	9.2.4.2.9	H6
17	1	9.4.2.2.5	H10
18	1	9.2.4.2.2, 9.2.4.3.1, 9.3.2.2.5/7	H7, H9
19	1	9.2.4.2.4	H6, H9
20	1	9.3.1.3.5/9.3.2.2.4	H7, H9

Section I Part B

Question	Marks	Content	Syllabus outcomes
21 (a)	2	9.2.3.2.1	H6, H9
21 (b)	2	9.2.1.3.3	H6, H9, H14
22 (a)	3	9.3.5.3.1	H7, H9
22 (b)	3	9.3.2.2.4, 9.3.2.3.2	H7, H9, H13
23 (a)	2	9.4.1.2.9	H7, H9
23 (b)	3	9.4.1.2/2/3/5/9	H9
24 (a)	2	9.3.3.3.4	H7, H9
24 (b)	3	9.3.1.2.2, 9.3.1.3.1	H9, H14
25 (a)	2	9.4.4.3.5	H4, H7
25 (b)	4	9.2.3.2.1, 9.4.1.2.6/7, 9.2.1.2.1	H9
26	4	9.4.4.2.3/4/5/6	H7, H9
27 (a)	3	9.2.2.3.1/2	H9, H11
27 (b)	3	9.2.2.3.1/2	H9, H14

Question	Marks	Content	Syllabus outcomes
28 (a)	2	9.2.1.2.3	H7, H9, H14
28 (b)	3	9.2.2.3.3	H7, H9
29 (a)	4	9.4.2.3.3, 9.4.3.3.4	H7, H10
29 (b)	4	9.4.3.2.2/5/6/8, 9.4.3.3.3	H3, H7
30	6	9.3.3.2.4, 9.3.4.2.4, 9.3.4.3.4	H7, H9

Section II

Question	Marks	Content	Syllabus outcomes
Question 31		Geophysics	
(a) (i)	2	9.5.1.2.2	H7
(a) (ii)	3	9.5.1.2.2	H3
(b) (i)	2	9.5.2.2.2	H4
(b) (ii)	3	9.5.2.3.4	H14
(c)	4	9.5.2.3.2, 9.5.1.3.1	H7, H9
(d)	4	9.5.3.2.1/2/5	H7, H8
(e)	7	9.5.4.2.3, 9.5.4.3.2	H9
Question 32		Medical Physics	
(a) (i)	2	9.6.4.3.2	H4
(a) (ii)	3	9.6.4.2.4/5	H7, H9
(b) (i)	2	9.6.1.2.4, 9.6.1.2.6	H7, H14
(b) (ii)	3	9.6.1.2.5	H7, H8
(c)	4	9.6.3.2.1/4	H7
(d)	4	9.6.2.2.3/4, 9.6.2.3.2	H7, H8
(e)	7	9.6.2.2.3/4, 9.6.2.3.2	H7, H8
Question 33		Astrophysics	
(a) (i)	2	9.7.2.2.1	H8, H10, H13
(a) (ii)	3	9.7.1.2.4, 9.7.2.2.3	H8, H10
(b) (i)	2	9.7.5.2.1/3, 9.7.5.3.1	H8
(b) (ii)	3	9.7.4.3.1, 9.7.5.2.4	H8, H13, H14
(c)	4	9.7.6.2.3, 9.7.6.3.2	H7, H10
(d)	4	9.7.4.2.4	H3, H8, H10
(e)	7	9.7.3.2.1/5, 9.7.3.3.1	H8, H10, H11, H14
Question 34		From Quanta to Quarks	
(a) (i)	2	9.8.1.2.6	H2, H10
(a) (ii)	3	9.8.1.2.2	H1, H10
(b) (i)	2	9.8.3.2.2	H1, H6, H10
(b) (ii)	3	9.8.3.3.2	H7, H10
(c)	4	9.8.3.2.6	H1, H7, H10
(d)	4	9.8.3.3.1	H10

Question	Marks	Content	Syllabus outcomes
(e)	7	9.8.3.2.1/7/8, 9.8.4.2.5	H1, H9, H10
Question 35		The Age of Silicon	
(a) (i)	2	9.9.2.2.1	H3, H7
(a) (ii)	3	9.9.1.3.2	H3
(b) (i)	2	9.9.6.2.3/7/9	H13
(b) (ii)	3	9.9.6.2.2/6/8, 9.9.6.3.1/2	H7
(c)	4	9.9.7.2.1/2, 9.9.7.3.1	H3
(d)	4	9.9.4.2.1/2, 9.9.4.3.1	H7
(e)	7	9.9.2.5/6, 9.9.2.3.3, 9.9.3.2.5/6, 9.9.3.3.1/2, 9.9.5.2.1/2, 9.9.6.2.5	H9, H3